PREDICTING THE RESULT OF FOOTBALL MATCHES

A DISSERTATION SUBMITTED TO MANCHESTER METROPOLITAN UNIVERSITY
FOR THE DEGREE OF MASTER OF SCIENCE
IN THE FACULTY OF SCIENCE AND ENGINEERING



2023

By
Chinthaka Gamage
Department of Computing and Mathematics

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Abstract

The game of football is rooted back to the ancient Egyptian era, and its popularity could be imagined with the continued expansion of up to 14.56 million viewers on the most popular English Premier League in the 2017/18 season, as per ESPN records (Alfredo and Isa 2019). Since 1930, huge attention has been drawn by sports lovers, performance evaluators, scholars, and punters, whilst the prediction of the match results has been pushed out as an intriguing research problem (Rodrigues et al. 2022). Software solutions including machine learning algorithms have been used profoundly to address the empirical gaps that had existed, in improving the prediction accuracy of a game of a stochastic nature, over traditional analysis. However, with the game's unpredictability, the accurate predictions of match results demand continual improvements. Accordingly over 50 publications were referred including journal articles to get the wisdom and insight needed to make the months of study, productive. Capturing a versatile dataset with 45952 observations, the most influential subset filtering 138 features, and selecting the appropriate and diverse combination of algorithms with deep learning and ensemble models, supported by the expert domain knowledge gained, eventually revealed this robust model. The features were integrated with team form, goals conceded, home, and away team factors, and predictions by bookmakers through betting odds. A sound methodology followed by iteratively modifying the design through a rigorous process of trials immensely improved the model accuracy against the skewed indices primarily observed in draw predictions. ELO rating and data clustering were investigated in bridging gaps. The SMOTE sampling along with improvised feature engineering works revealed marked performance indices. A rigorous process of hyperparameter tuning optimized the model performances as led by the Random Forest Classifier with 0.66 prediction accuracy. The model was evaluated through a few directions to ensure model robustness, prior to preserving the application for guarantee match results prediction.

Declaration

No part of this project has been submitted in support of an application for any other degree or qualification at this or any other institute of learning. Apart from those parts of the project containing citations to the work of others, this project is my own unaided work. This work has been carried out in accordance with the Manchester Metropolitan University research ethics procedures, and has received ethical approval number Your EthOS Number:57278.

Signed: 

Date: 29/092/023

Acknowledgements

It is with great pleasure that extend my heartiest regards to the stakeholders of this project since it would never be a manageable situation for me, despite their contribution. My beloved wife got the presidency as she relieved me from all the burdens on the domestic front, bearing all hardships. My project supervisor, Dr. Xia Cui has been a real source of inspiration by professionally showing me the corners and encouraging me with vital directions throughout. The support extended by batch mates sharing knowledge cannot be ignored.

Abbreviations

EUFA	European Union Football Association
EPL	English Premier League
France	French League 1
Spain	Spanish La Liga
Italy	Italian Seria A
Germany	German Bundesliga
FIFA	Federation Internationale de Football Association
EPL	English Premier League
NB	Naive Bayes
GNB	Gaussian naive Bayes
SVM	Support Vector Machine
KNN	K Nearest Neighbour
BLR	Binomial Logistic Regression
RFR	Random Forest Regressor
RFC	Random Forest Classifier
GBR	Gradient Boost Regressor
MLR	Multinomial Linear Regression
GB	Gradient Boost
MLP	Multi-layer Perceptron
ANN	Artificial Neural Network
DNN	Deep Neural Network
MNN	Multi-class Neural Network
VC	Voting Classifier
VE	Voting Ensemble
PCA	Principle Component Analysis
QBC	Query By Committee

RFE	Recursive Feature Elimination
DBSCAN	Density-Based Spatial Clustering of Application with Noise
GMM	Gaussian Mixture Model
FTR	Full Time Result
FTHG	Full Time Home Goal
FTAG	Full Time Away Goal
HTHG	Half Time Home Goal
HTHG	Half Time Away Goal
HS	Home Team Shots
AS	Away Team Shots
HST	Home Team Shots on Target
AST	Away Team Shots on Target
HF	Home Team Fouls Committed
AF	Away Team Fouls Committed
HC	Home Team Corners
AC	Away Team Corners
HY	Home Team Yellow Cards
AY	Away Team Yellow Cards
HR	Home Team Red Cards
AR	Away Team Red Cards
PCA	Principle Component Analysis
SMOTE	Synthetic Minority Over-sampling
ADASYN	Adaptive Synthetic Sampling
ENN	Edited Nearest Neighbours
KPI	Key Performance Indicator
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
TPR	True Positive Rate
FPR	False Positive Rate
ROC	Receiver Operating Characteristic Curve
AUC	Area Under the Curve
SHAP	SHapley Additive exPlanations

Chapter 1

INTRODUCTION

1.1 Background

Sports activities had emerged to relieve the internal aggression of the communities in a peaceful manner and hence showed great excitement among communities in modern sports. The game of football is rooted back to the era of ancient Egypt where the ball was made out of linen fabrics. The most played and most watched sport on earth is football (Rudrapal et al. 2020). Alfredo(2019) evident expansion with 14.56 million viewers on the most popular English Premier League in the 2017/18 season, as per the Entertainment and Sports Programming Network (ESPN) records. A great attention was drawn among sports lovers, punters, and scholars, Since 1930. The prediction of the match results eventually has become an intriguing research problem, with the eagerness of the sports clubs to make decisions on future games, demanding a robust mechanism for the match results prediction, against the traditional statistical approach persisted. The predictions over the unpredictability have been always a challenging task with the complexity of the numbers and the nature of contributory factors. Software solutions including machine learning (ML) algorithms have been used to address the empirical gaps progressively elevating the prediction accuracy of match results (Buyrukoğlu and Savaş 2023; Bunker et al. 2022; Baboota and Harleen Kaur 2019; Razali et al. 2017). Yet, due to the subjectivity of accurate predictions of match results has been a never-ending process with the intensity of the competition with the stakeholders involved and the daily sophistication of the concerned software. The pieces of literature were referred, to in view of getting the wisdom towards the project objectives. In 1992, the American Statistical Association created a section

dedicated to sports statistics and the use of statistical techniques in sports whilst data-driven decision-making came to media attention in 2002 (Ievoli, Palazzo, and Ragozini 2021). Eventually, the prediction through ML algorithms was first applied in the International Federation of Football (FIFA) in 2014, and even it was extended to other sports as evidenced by the national scouting team Germany's study of the opposition team in decision making. Bunker(2022) expresses the dominance of football in terms of the algorithm utility, during the period from 1996 to 2019. Football governing body, FIFA estimated at the turn of in 21st century, there would be approximately 250 million football players and over 1.3 billion people interested in football, as a game that extends with great unpredictability of 90 minutes. Further, the English Premier League (EPL) is the most watched football league in the world with 4.7 billion viewers (Rudrapal et al. 2020; Rodrigues et al. 2022; Azeman et al. 2021).

The ML algorithms are primarily bi-fold as supervised and non-supervised where the former uses the input data for the paired predictions whilst the latter predicts the trends and patterns of the provided data itself. Hence, the supervised ML model has opted for the task and use for the prediction of regression and classification tasks based on the numerical and categorical target predictions. The football result (Full-Time Result) is a categorical target attribute and hence classification algorithms could be targeted. However, the regression work even could be done, in case the numerical target like full-time goals is selected. Yet, prediction has never been an easier task and extracting the most influential and contributory features plays the most important role in terms of prediction accuracy. Bunker(2022) states that the process demands feature engineering work in view of getting the precise subset of data, in elevating the predictability. The selection of a few appropriate algorithms would harness many gaps in finding the best model for the task and over the empirical evidence and the domain intuition, the k-nearest Neighbour (k-NN), Random Forest (RFC), Gradient Boosting Classifier(GBC) and Multi-layer Perceptron (MLP), and Voting Ensemble (VE) classification models were used, compared to the observed performance of the Logistic regression, support vector machine(SVM), Gaussian naive Bayes (GNB) models came a cross, as per the literature referred (Baboota and Harleen Kaur 2019; Z. e. Al. 2019; Chinwe Peace 2014; Harmandeep Kaur and S. Jain 2018; Pugsee and Pattawong 2019; L. e. Al. 2012; Alfredo and Isa 2019; Guan and X. Wang 2022; Maanijou et al. 2019). The evaluation metrics such as Confusion matric, cross-validation, ROC-AUC curves, and True vs. Predicted analysis were used to reveal the best robust model.

1.2 Problem Assessment

With the emergence of artificial Intelligence (AI), smart predictions through ML have been evident with great credentials to form strategies effectively over future games (Buyrukoğlu and Savaş 2023; Bunker et al. 2022; Baboota and Harleen Kaur 2019; Razali et al. 2017). Alfredo(2019)states that the game football has been the most popular sports activity and match result predictions have been challenged over the chance events induced by diverse contributory factors beyond the talents (Berrar et al. 2019). Further, the prediction of the football match results has become an endless process over the complex context with the mediation of the investors and rapid development of software in predictions. Berrar(2019), added that a complex mix of factors and chance events play a major role in determining the final outcome of a football match result, as more than 85 per cent of matches have ended as a draw otherwise a win by only two or fewer goals. The situation led to the building of a robust model for football match result prediction, whilst using the same in fulfilment of the mandatory compliance, embedded with the master study following.

1.3 Aim and Objectives

This project is on predicting the results of football matches by (ML) models to make decisions on future games. To meet the aim, the below objectives are set.

1. Collect an appropriate dataset and understand the tasks and features in it.
2. Process the data in view of the model application.
3. Selecting potentially contributed features for the match result.
4. Train ML models for the prediction of the football match results, using the selected features of the dataset.
5. Validate the prediction with the actual football match results.
6. Record and compile the report that reflects the project objectives and learning outcomes.

Structure Accordingly, the study was segmented into a few chapters from the introduction. Chapter (2) would be the Literature Review, which critically evaluates the past empirical evidence pertaining the football match prediction. Chapter 3 is on understanding the data. Methodology and Model Design are followed. Chapter 5, the Implementation and Results would eventually explore the model outcomes. The model Evaluation will be the subsequent chapter where the work done will be critically examined with respect to the set aims and objectives. Chapter 7, Conclusion would entail the summary of work including future work which needs to be addressed in future studies. The references will be attached finally.

1.4 Tools and Timeline

The project Gantt Chart is included for the ease of assimilation of the proceedings (figure1.1).



Figure 1.1: The Gantt Chart of the project proceeding

Chapter Summary The chapter introduces the background of the study which is rooted in the Egyptian era, its progression and the necessity of using ML algorithms in results prediction. Accordingly, this study was selected with aim of predicting football match results through 9 chapters.

Chapter 2

THE LITERATURE REVIEW

2.1 General

In this chapter, the empirical evidence pertaining to the origin of football and its expansion, and football match results prediction through ML algorithms, evidence pertaining to preprocessing and post-processing work over the course were enveloped. The attention was drawn to dataset, feature and algorithm selection, feature engineering work, applying ELO rating, Clustering and evaluation metrics.

Football referred to as soccer, is the most popular sport in the world and would serve the well-being of people immensely. EPL, French League 1(France), Spanish La Liga (Spain), Italian Seria A (Italy) and German Bundesliga (Germany) are well-known famous premier League tournaments in Europe and even in the world. The dominance of the game of football during the period from 1996 to 2019 is evident (figure2.1). With the greatest unpredictability observed in the 90 minutes of play, the emergence of AI tools has made a huge impact in predicting match results, elevating the excitement among sports lovers and in modern sports, investors. Eventually, the ML predictions were used in many games (Buyrukoğlu and Savaş 2023; Bunker et al. 2022; Baboota and Harleen Kaur 2019; Razali et al. 2017; Athish, Rajeswari, and Nandha 2023) and are in use extensively today to serve the dynamic need. Buyrukoglu(2023) claimed that the ML methods have been applied in quite a diverse disciplines over the high performance shown in the recent past. Hence, researchers have used the algorithms in football as well in different predictions such as estimation of affecting factors, prediction of league standings, and analysis of the performance of players, despite the dominance in football match results prediction.

The predictions were mainly being done by two means basically; result-based and goal base where match results classes and number of goals scored are revealed (Baboota and Harleen Kaur 2019). Razali(2017) used primarily the Bayesian Network(BN), a time series approach for 3 seasons; 2010/11, 2011/12, and 2012/13 of the EPL that involved 20 teams with 380 matches, based on objective and subjective features including; team strength, team form, psychological impact and fatigue. Bunker(2022), used the Model accuracy (the number of correct predictions), as the primary metric in the prediction processes. The K-fold cross-validation (CV) was used for testing the accuracy to secure marked figures of 75 per cent, over decision tree learner, k-Nearest Neighbors (KNN). 90 per cent data was used for training the model. This was a marked score over the majority of other studies mentioned.

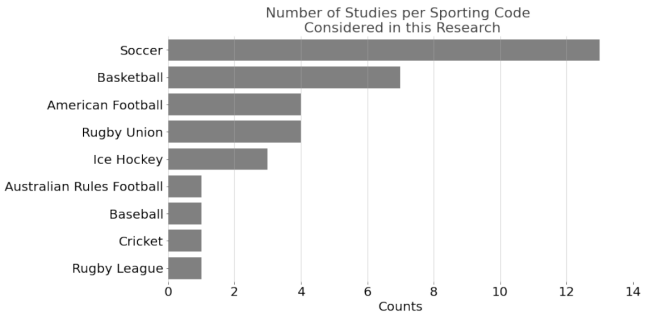


Figure 2.1: Presidency on football, **Bunker et al.(2022)**

2.2 ML Algorithms in use for the football results prediction

”ML is referred to as a scientific and systematic domain, that critically explores the creation, study and overall application of a broad-based algorithm that can learn from datasets or model datasets” (Igiri et al. 2015; Buyrukoğlu and Savaş 2023). Buyrukoglu(2023) emphasizes the importance of using ML algorithms in results prediction, estimation of factors affecting match results in many disciplines, over the course of ability in reproducing, productivity and more importantly robustness, ahead of traditional means. ML is twofold as Supervise and Non-supervise learning where maps input data to project output in former as a classifying work in case of categorical target, for example, Full-time Result (FTR) and a regression outcome where the target is a numerical feature, for example, Full-time Home Goals(FTHG). In contrast, unsupervised learning is used to discover patterns and structures of data itself and does not reveal explicit labels or targets. Based on the application, the right fold is chosen where both extend unique advantages in the analysis and a hybrid method even could be used. In addition, reinforcement learning is used for interacting with an active environment without a trainer. The prediction of the football match results and the choice of the algorithms play a crucial role in model prediction (Buyrukoğlu and Savaş 2023; Bunker et al. 2022; Baboota and Harleen Kaur 2019; Razali et al. 2017). In this study, a target is given as a prediction of match results and hence, the supervised learning algorithms would be focused. In the case of predicting the match result, it is very clear that the classification of ML models could be used. The literature showcases the applicability of a few important algorithms for the study of football match results prediction where the actual need is to have three outcomes Home win, Draw and away win, and the accuracy of prediction is a concerned factor, hence algorithms listed below have drawn much attention, based on the stated empirical evidence.

2.2.1 Gaussian Naïve Bayes model(GNB)

The GNB model, a simple, computationally efficient algorithm that has shown suitability to handle large datasets and even smaller ones with real-time applications, frequently served as a baseline model to benchmark more complex models (Baboota and Harleen Kaur 2019). Baboota and H. Kaur (2019) emphasise the term Naïve means a strong assumption of independence, that is features do not have a correlation with each other. Further, it assumes that the data associated with each class is distributed as per the probability density function, which provides a probabilistic framework for classification as calculates the probabilities of different classes, and features, and allows intuitive interpretation towards the prediction of the results. It uses conditional probability to compute the posterior probabilities of a data instance belonging to a particular class. However, the model demands continuous numerical features and is said to be robust against irrelevant features, with the assumption of feature independence. The model interpretations are eased out with the predictions made, based on the probabilities, despite the limitations imposed in scenarios where the feature correlations exist and adherence to the data classes. The model was accordingly tested in this process and scores were below even didn't reach 60 per cent compared to the 70s and 80s marked in other models, finally forced to be removed from this study. The results were further optimized with 85.43 per cent accuracy in the latest football prediction using the GNB (Athish, Rajeswari, and Nandha 2023). However, the algorithm was least performing in this study forcing it to be removed from the process to avoid computational complexity, as referred by (figure2.2).

2018 World Cup Season	*#Wins*	*Accuracy*
Whatifsports.com	36	56.25%
Christodoulou's first approach	21	65.62%
Christodoulou's Second approach	44	68.75%
Neural Network	45	70.31%
Support Vector Machine	48	75.00%
Decision Tree	51	79.81%
Gaussian Naïve Bayes	**54**	**85.43%**

Figure 2.2: GNB Performance, **Babboota(2019)**

2.2.2 Support Vector Machine (SVM)

A tool identified as better performing in high-dimensional feature space that outlines the most appropriate hyperplane over the training data input. SVMs offer a universal mechanism to fit the surface of the hyperplane using a kernel function and use the least risk of structuring, revealing better generalization ability. During the training process, the SVMs are given inputs (such as linear, polynomial, or Sigmoid functions), which pick support vectors along the surface of this function (Z. e. Al. 2019). SVMs do not require a reduction in the number of input features in order to avoid overfitting which is an apparent advantage in applications like intrusion detection. The number of free parameters used in SVMs depends on the margin that separates the data points rather than the number of input features. However, a study achieved 53.3 per cent accuracy and claimed that pure SVM may not be good enough for the football match prediction (Chinwe Peace 2014; Harmandeep Kaur and S. Jain 2018). A trial application in this study confirmed the lower accuracy and hence decided not to proceed with the model. The low predicted chance of generalisation errors is a key benefit of SVMs (figure2.3), however, the use of the model would be determined based on the performance indices.

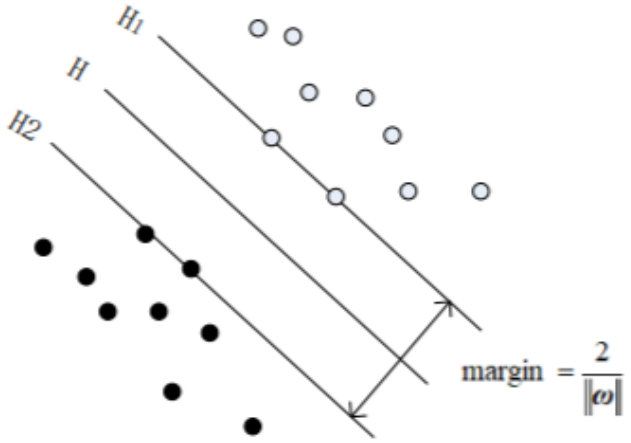


Figure 2.3: SVM Architecture, **Zhang et al.(2019)**

2.2.3 k-Nearest Neighbours (kNN) Classifier

The kNN approach does not provide a classifier or model, however, it classifies samples by comparing them to the k-closest data points in a given dataset. In predicting contest outcomes using data from in-play scenarios, kNN techniques perform quite well (Ren and Susnjak 2022). In the case of the prediction model, the technique must take into account all of the sample's data. In competitive sports, using this method to predict match results carries a significant risk of overfitting. However, it is yet unknown how effectively the kNN approach can work for pairings that are difficult to forecast. A comparative analysis of kNN and ANN techniques made by Igiri et al.(2015) on EPL data shows a moderate accuracy of 53.33 per cent and claims better results in sports analysis with appropriate data. The study of "Incorporating domain knowledge in ML for soccer outcome prediction was able to top rank the kNN model, with newly introduced features "Recency feature extraction and rating feature learning", subjected to domain knowledge (Berrar et al. 2019). The study on "Automated web usage data mining and recommendation system using kNN emphasises that the classification method" gains over 70 per cent, apart from the transparency, straightforwardness, consistency and the simplicity shown (Adeniyi, Wei, and Yongquan 2016).

2.2.4 Random Forest Classifier(RFC)

Liu et al.(2012) express that, RFC is a combination of a series of tree structure classifiers with extended advantages and hence that has been used extensively for the predictions, (figure2.4). It is explained that the model is embedded with multiple decision trees that depend on independent sampling with even randomness in the selection of data and polled to return the category with the most votes with scores attained by each variable based on the importance and assessing the weighting given through each variable to the classification (Ren and Susnjak 2022; Hucaljuk and Rakipović 2011). It is said that the model is more robust in terms of overfitting compared to the decision tree algorithms. The correlation among features has been an issue in machine learning predictions and the different criteria in classification matters in the random forest is advantageous. A study was carried out on " Football Match Results prediction using RFC", with three seasonal EPL match data and convinced that they have marked over 70 per cent of both accuracy and precision in the prediction task (Pugsee and Pattawong 2019). During the model training, the training data was used to generate an

individual classification model by the decision tree, which will display the data classification as a tree and identify the data in nodes with impurity data. However, it is claimed that the variation of the classification depends on the respective data set and design attributes to avoid overfitting issues. In the application, the knowledge value for each attribute was calculated by using the impurity measurement entropy data and higher-ranked values were used to divide the data in building the model with combined multiple decision trees. Further, it is stated that the RF shows high classification accuracy whilst tolerating the outliers and the noises and never gets overfit (L. e. Al. 2012).

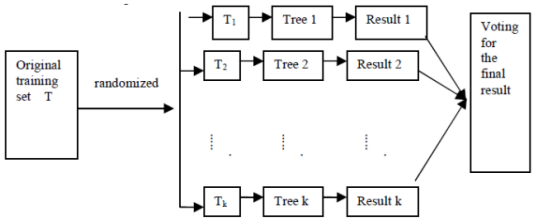


Figure 2.4: Random Forest Schematic, **Liu et al.(2012)**

2.2.5 Gradient Boost

This is a bit identical to the Random forest as tree-based. With the Gradient boosting, many weak models are created iterative way, and the predictions from each weak model are then added together (Ren and Susnjak 2022). The weights of the incorrectly split samples grow with each residual calculation, while those of the split pairs shrink to zero, improving generalisation performance. Yet, the Match prediction gradient boost approaches have frequently failed to yield the best models in prior studies[1]. However, it is impossible to categorically refute the method's ability to anticipate outcomes in hard-to-predict matches. (Alfredo and Isa 2019) stresses in his article published on "Football match Prediction with Tree-Based Model Classifications" (that is Random Forest and the Gradient Boosting Classifiers) are not good enough in predicting football match results despite the accuracy score gained of 67.89 per cent. It was done for the 10 seasons from 2007/2008 to 2016/2017 on EPL data that used 15 initial features for the prediction. However, it is evident that even the model here scored over 75 per cent accuracy, despite being the lowest among them.

2.2.6 Artificial Neural Network(ANN)

In contrast to the above models, a study was done in ”Optimization analysis of football match prediction model based on the neural network”, that uses the deep learning process (Guan and X. Wang 2022). The architecture is completely different compared to the algorithm discussed and needs more elaboration. A pertinent study has used the BP neural network, a powerful tool in processing non-linear data with grey fuzzy, grey extreme and grey fuzzy extreme combined models for the prediction. Eventually, the neural network model was tested, in terms of prediction accuracy, speed of data processing and data transmission accuracy. The model uses the quantum particle swam algorithm to contain prediction errors below 1 per cent that pave the high learning efficiency and the convergence speed and attain 95 per cent accuracy in a gesture recognition analysis, despite the drawback in learning rate adaptability. In case of understanding, an Extreme learning machine algorithm could be used contained with a single hidden layer feedforward neural network and used a simple least square method to determine the best output weights overcoming to input weight, hidden layer deviation and the determination of output weights, compared to traditional iterative learning modes. It is pertinent to have a sense of the network to gain an insight into the process through the below diagram with the extreme learning machine structure (figure2.5). They emphasise the generalization performance of the model and the ability of faster training capability and are left with selecting the appropriate activation function for a user where Sigmoid and the ReLu are more popular. Three min layers could be identified including input and output as hidden layers for processing.

The training flow chart of the BP neural network would induce better insight into the functionality of the model (figure2.6). Accordingly, reverse calculation of training error and modifying the weights to calculate the global error, elevate the error accuracy, thereby, the model showing a unique structure for modelling to realize the competitive edge. The algorithm flowchart of the grey fuzzy neural network would make better arousal and sense for an ordinal reader (figure2.7). A comparative analysis of k-NN and ANN techniques made on EPL data by Igiri at al.(2015) showed the formidable level of accuracy of 70 per cent with the polynomial label. Buyrukoğlu and Savaş(2023), used the Deep Neural Network (DNN) for the latest prediction study and reserved over 72 per cent of Accuracy, Precision, Recall and F1-score with FIFA’19 dataset (figure2.8). In another study, the Multi-layer Perceptron (MLP) classifier model was used to predict the football match results in 2020 along with SVM, GNB and RF and MLP secured the optimum figures in terms of Accuracy of 74 per cent, Sensitivity

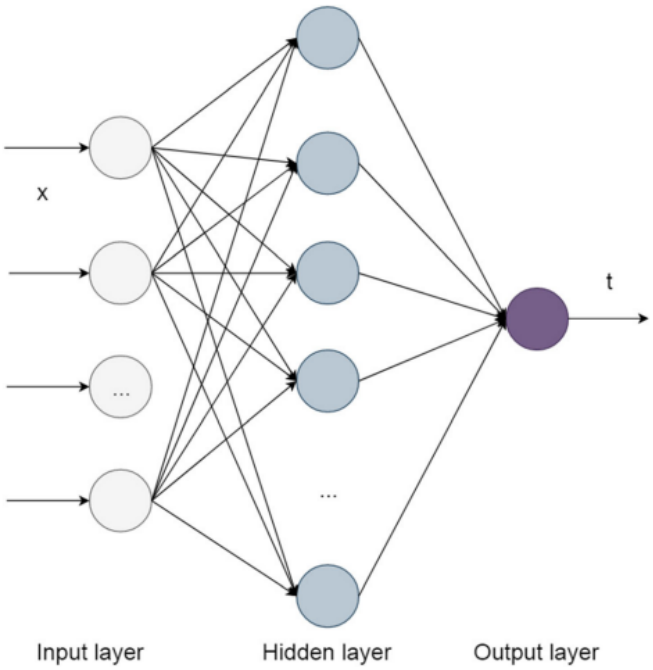


Figure 2.5: Neural Network Architecture, **Guan and Wang(2022)**

of 72 per cent, Specificity of 75 per cent and F1-score of 71 per cent (Rudrapal et al. 2020). The figures could be tabulated with marked performances, compared to the figures of the other 3 models. In an article published on ”Novel method for prediction of Euro League games results using hybrid feature extraction and machine learning techniques” by Ballı and Özdemir(2021) on basketball results prediction stated achievement of 98.90 per cent prediction success where they had used the 2012-2013 and 2016-2017 match seasons data for the predictions.

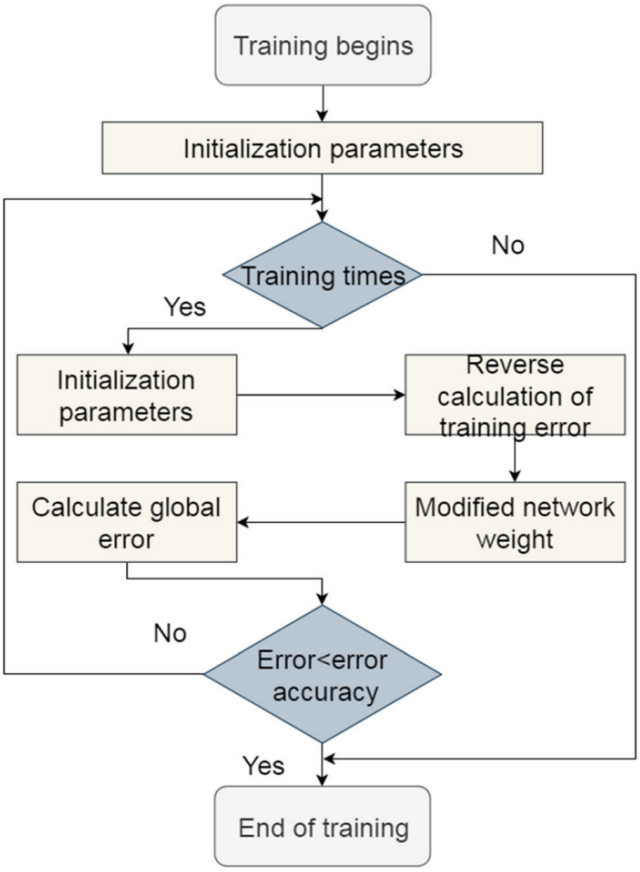


Figure 2.6: BP Neural Network Flowchart, **Guan and Wang(2022)**

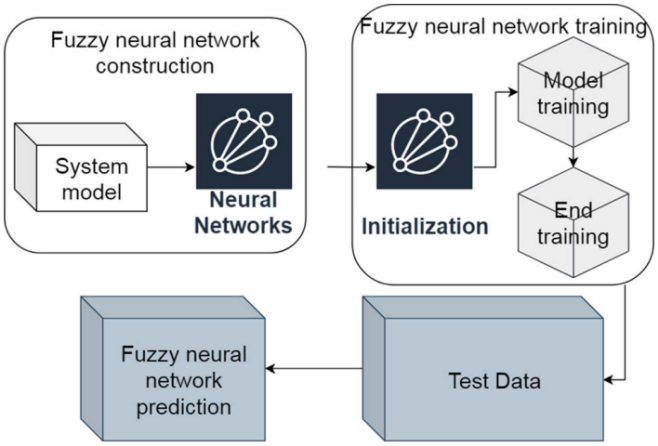


Figure 2.7: Algorithm Flowchart of grey hybrid fuzzy neural network, **Guan and Wang(2022)**

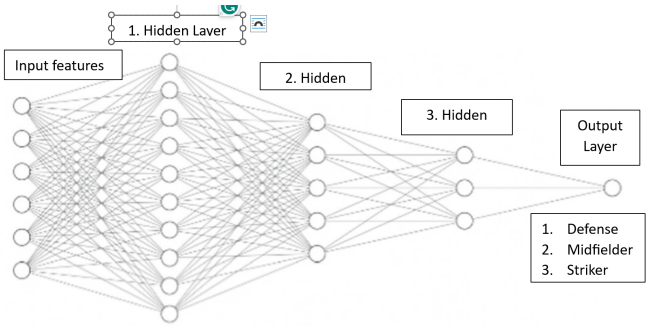


Figure 2.8: Structure of Deep Neural Network, **Buyrukoglu(2023)**

2.2.7 Voter Ensemble model

The model is a multiple classification type, compared to other algorithms as this combines them to a single decision and connects them to a single result. The required conditions for combining different methods are accuracy and diversity (Maanijou et al. 2019). Diverse methods are those that make different results on new points so that their inputs can be combined to reveal better classifiers. The model was used in the prediction of basketball results (Ballı and Özdemir 2021). It emphasised a series of decisions as Majority of voting, Average probabilities, product of probabilities, maximum probability, Minimum probability, and Medium. A schematic diagram for ensemble learning to combine classification algorithms could be provided (figure2.9). Accordingly, the model predictions are calculated separately to harness a unique output after a vote, which results in a powerful outcome in almost all cases.

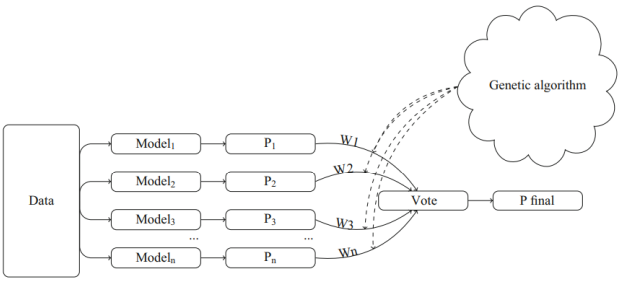


Figure 2.9: Schematic of Ensemble Learning, **Maanijou and Mirroshandel(2019)**

2.3 Pertaining Other Empirical Evidence

The popular Bayesian network was used in 2011 for a prediction of football match results of Tottenham Hotspur in the 1995/96 and 1996/97 seasons of the Premier League based on 3 classification attributes (Home Win, Away Win, Draw) along with naive Bayes, Decision Tree group (MC4), kNN, Logist Boost, Random Forest, and ANN classifiers with selected 20 features such as Home court advantage, overall and individual quality of the team and player respectively, current form, number of injured players and average number of goals scored (Hucaljuk and Rakipović 2011; Ren and Susnjak 2022). The training and validation data are used with different rounds and found optimum results when increasing the training set from 3 to 4 rounds with random forest. However, the study reveals that further searching is needed on the inclusion of the form of the player with the larger dataset for optimization.

A study was carried out with different features; players' combination, form, teamwork, extra time, or substitute player factor, and injuries affected, advantages of venue, crowd support and the known environment with binary classification as home and away. In addition, team rating, market value, EPL time point, per cent performance, attack point, and defence point are considered (Rudrapal et al. 2020). The player rating and the market value had been defined as the player's overall rating on aggression and the form of the player, dependability, and the number of seasons left in the career, respectively. 70 per cent of the data being used as the training data and the recorded accuracy of 74 per cent, Sensitivity of 71 per cent and an F1-score of 71 per cent whilst RF secured the next layer with an accuracy of 73 per cent and 66 per cent F1-Score. However, the task only considered starting 11 players for the study and proposed considering all players including substitutes, coach ratings, reserve player strength, and future studies to elevate the prediction accuracy. A marked performance was evident with MLP of 83 per cent accuracy and equivalent other indices.

A major study was done emphasises a Multi-class Decision Forest (MDF) Multi-class Neural Network (MNN) application of 380 matches played between 20 teams in the 2005/06 EPL seasons with 19 selected features out of 65 (Azeman et al. 2021). The prediction results show the superiority of MLP with 88, and 92 per cent overall, average accuracy respectively, over MNN with computational expensiveness. The reflected future proposals envisage using more algorithms on different leagues to make the model more robust. In addition, the sophistication of the Random Forest model with broad dimensions with tubular and categorical values paved the way to aggregate the errors to reveal elevated accuracy, despite the capacity of neural networks.

A betting work developed through the ML methodology in support of match results prediction as an iconic tool to the business's commercial needs of investors (Rodrigues et al. 2022). The EPL spanned for five seasons from 2013/14 to 2018/19, with 1900 matches over a quieter new bunch of features; individual team shots made, number of corners, fouls, yellow cards, red cards, odds for each match, and results, match referee, pass accuracy, agility, reaction, and aggression of the individual players, overall rating, rating of team attack and defence were the selected features. The data in the entire range was considered to avoid the gaps at the beginning and end of the seasons over the issues with the form and the fatigue factor, respectively. The data span avoided the model overfitting by taking 1520 games for training. The Naïve Bayes (NB), KNN, RF, SVM, C5.0 (Decision Tree), Xgboost, Multinomial Linear Regression (MLR), and ANN where the highest accuracy marked by SVM as 61 per cent securing betting profit of 95.06 Euros. The data had been normalised before testing to avoid the effect of large variations in the values. The project limitations were identified as the forecasting model to be integrated into a decision support system that accesses the risk of bets based on the probability of occurrences of the forecast model results.

In addition, features such as ball possession, completed passes, diameter and betweenness, offensive actions, finalization, centralization and clustering are not always significantly informative. One study was carried out based on the European Union Football Association(EUFA) that spanned out 96 matches among 32 teams in the 2017/18 seasons (Ievoli, Palazzo, and Ragozini 2021). Some approaches considered the ball possession, others multivariate analysis such as Principle Component Analysis (PCA) and clustering to distinguish players' market to determine how much either team's reputation or individual skills affect the market value. However, not all methods consider the strength of the relationship between players, the type of interactions among them both at the micro level(players) and macro level(team) and the global team organization. Passing network properties provides insights into the team's style of play, strategy, and coordination on the field. This enables to identify key players, understand their roles, and assess the effectiveness of the passing strategies to make informed decisions to improve team performance. Binomial Logistic Regression (BLR) and a few comparable algorithms modelled the probability of winning the game, showing how passing network information could be used to improve model performance, in terms of Sensitivity, Specificity, accuracy, and Area Under the Curve (AUC). The BLR showed the best performance. Rodrigues et al.(2022) found that the most influenced prediction features as home and away defence, however, their predictions were done

only based on binary factors such as win and lose excluding the draws.

The prediction of the match results of Barclay's Premier League season 2010/2016 to determine the significant features for the home and away wins of the matches (Prasetio and Harlili 2016). The sofia.com data gathered from video games had been used, additionally in modelling. The offensive and defensive variables of Home Teams and Away Teams were experimented with by altering seasons of training data and revealed a prediction accuracy of 70 per cent. The selected features were home advantage, recent team performance, and team strength along with some others. The training was done to estimate a coefficient of variable, varying the training data composition whilst testing was done using the data in 2015/2016. The predictions were done over the coefficient and compared with real results to gain accuracy iteratively, varying the training composition. The recorded accuracy was 70 per cent using the whole range of data where the home defence and away defence were the selected features, which is an optimum result with a few variables, compared to the predecessor with 51 per cent accuracy. However, they concluded by proposing the need for another analysis of constant features, despite the lower accuracy marked, compared to the moderate figures and the elevated indices of this study.

ML applications are ubiquitous with result prediction and betting, currently (Carloni et al. 2021). In the study, it was applied LR, KNN, SVM, Naïve Bayes, and Random Forest (RF) algorithms along with ANN compared the prediction accuracy in the evaluation and ANN revealed the better outcome. Their studies revealed drawbacks due to the inadequacy of data set size and the need to use player-level statistics and more advanced ML models. Web scraping technique was used to extract betting odds to have structured data and train the models selected and evaluated in the prediction. There were 36,461 matches involved in the study in the season from 2008/2009 until 2019/2020 with 47 features. Uni-variant feature selection, feature importance and correlation matrix had been used to reduce the number of features to 31. In identifying the best model, productivity was evaluated, reused the model for the incoming matches where ANN outperformed. Accordingly, ANN was applied based on the 80-20 rule in selecting training and testing data respectively. It was proposed to use a Recurrent Neural Network (RNN) instead of ANN as being found efficient in managing instances of different time intervals. However, the MLP routed from ANN was showing over 83 per cent with equivalent other indices pertaining to this study.

Another study was made on EPL data on "Football Match Winner Prediction" (Vaidya, Sanghavi, and Gevaria 2016). It was convinced that the way of sampling

makes a greater impact on the outcome and is empirically shown using random, Query By Committee (QBC), uncertainty and Density sampling, however, Random sampling has been more utilized over the drawbacks in other methods, despite issues in the utilization (Rojas et al. 2017).

Eventually, Baboota and Kaur(2019) confirmed the role of feature engineering in ensuring the quality of the workmanship as the most important aspect of prediction through their journal article (on Predictive analysis and modelling football results using the ML approach for EPL). That is realistic only with the precise subset of features through feature engineering. It highlighted the relative strength of the attack, midfield, overall rating and defence for both types of teams as collected from the rating statistics. The goal difference between the two teams, the corners, shots on target and the number of goals as important attributes in the study where corners show how well enforced whilst the shots on target represent the superiority of the team. The number of goals has a direct bearing on the final results of the match. The importance of using the recent data elevates the model's accuracy. Thus, the "streak" was introduced as a new feature with quantifiable insight and judgements with a given weight to outweigh the recent impact. The team form was additionally introduced with high and low form values. However, this study revealed, that the goal difference would lead to overfitting the model.

Feature Selection is very much essential to set prior to the model training process (Alfredo and Isa 2019). The study made through a deep learning approach revealed the selection of 40 features based on the empirical evidence (Rudrapal et al. 2020). It contained different classes of statistics over head-to-head match statistics, teams, and players. The matches had been classified into two main classes home and Away based on the locality of hosting since the same crucially affects the orientation and the crowd support for the prediction results. "a set of team-related features; attack point, midfield point, defence point, team rating, EPL times point, EPL streak point, EPL performance point, EPL point, standing points. The player rating features were separately taken as a player rating, player potential, market value, goalkeeper point, and form point difference". They added on some hidden features such as unfairness made by the referee, crowd support, and impact with the substitute players for better results.

Accordingly, it has served as an essential factor in terms of improving learning accuracy, learning efficiency and perceiving a better understanding of the correlations with the target feature and among each other, as it removes unnecessary, duplicated

and irrelevant data. It is a process of obtaining a selected subset of influential data. However, there would be a great propensity to overfit the model whilst adding noise, diluting the model focus. The process may involve data transformation and applying methodologies like Principle Component Analysis (PCA). The process entails with issues such as computational cost, and over-training of data (Cai et al. 2018; S B Kotsiantis 2014).

Baboota(2019) indicated that the feature engineering process reduced the features down to 33. The process was aided by dropping features with high correlation to reveal better performance. The Recursive Feature Elimination (RFE) was used with the GNB model since other models held inbuilt functions.

The predictions do not always become realistic. The website ”forbet” predicts the results based on the team strength, and goals scored but shows great unpredictability (0.2) with respect to the prediction done in the 2017/2018 seasons of EPL, demonstrating the risk of relying on these predictions (Rodrigues et al. 2022). However, the predictions were improved with the ”number of successful interceptions”, apart from the number of shots and goals scored as found during the root cause analysis of field prediction with respect to a game with Leicester City’s EPL. In addition, the goals conceded were identified among an influential 18 features considered, including the number of goals, shots, shots on target, number of corners, number of fouls, yellow cards, and red cards committed were considered up on average. However, to reduce the feature combinations, the ”backward” feature selection methodology of ”R software” was adopted. Furthermore, the study revealed the interaction of the B365D and B365A odds for the draw and win occurrence of the away team. In the case of classification, the feature with high correlation is to be removed to avoid the collinearity effect. The Boruta algorithm along with RF was used for the feature selection based on a heuristic variable selection algorithm (Rodrigues et al. 2022).

Further, it was focused on ”form, goals scored and conceded, shots ratio: where the form is given more insight and deeper meaning whilst the shot ratio and goals scored being combined to create a new feature (attacking quotient) to secure 47-50 per cent accuracy through a Voting algorithm with RFC and NB (Vaidya, Sanghavi, and Gevaria 2016). Further, they emphasise that the accuracy of the system would mainly depend on the features selected for the training and the models selected.

The dataset includes betting odds, including bookmaker odds and it is needed to verify the importance of using them for the prediction process. A journal on ”Using

bookmaker odds to predict the final result of football matches" published by (Odachowski and Grekow 2013), investigated the possibility of predicting how upcoming football matches will end based on the changes in bookmaker odds. I was evident by 70 per cent effectiveness in the existence of a relationship between changes in bookmaker odds values and the outcome of the football matches.

It talked about the importance of domain knowledge in securing elevated prediction results and they had developed two new feature engineering methods for the same as "Recency feature extraction and the Rating feature learning" (Berrar et al. 2019). Hence, it is paramount to have scientifically selected features and algorithms for the purpose of this project, based on empirical evidence to secure better indices.

Alfredo(2019) and Vaidya(2016) indicated that data was split as 80 (training data) and 20 (testing data) in the prediction of the football match results using tree-based classification secured over 64 per cent accuracy level. However, a study made to improve the accuracy for analyzing heart disease prediction based on the ensemble method divided the data set into 75 per cent training and 25 per cent testing sets and reached an accuracy of over 84 per cent (Gao et al. 2021).

Jain(2010) introduced clustering as a division of ML that comes under unsupervised segment use in grouping the data points based on given intrinsic characteristics, patterns, and similarities, which do not carry any predefined labels or categories. The main objective of the grouping is to discover the inherent structure within the data, to use for meaningful groupings. The structures are defined in a way that maximizes the intra-cluster similarity and minimizes inter-cluster similarity, where the data points within the same cluster are more similar to each other compared to those in different clusters.

It was highlighted how important the model selection is in getting the better prediction needed over football match results (Vaidya, Sanghavi, and Gevaria 2016). A recent study with satisfactory results (RF; 65 per cent accuracy and profit margin of 26.74) revealed. Logistic Regression (Lm), SVM, Naive Bayes(NB), KNN, RFR, GBR and ANN took the presidency in terms of utility. The model accuracy was primarily used to verify the strength of the prediction along with profit margins, percentages of home and away wins and draws (Rodrigues et al. 2022). The ternary classification had been reduced to binary in many classification problems. However, the Gaussian naive Bayes, SVM, RF and GB methods used in the study made by Baboota and Kaur(2019) achieved a model ranking of 0.2156 through GB for the EPL aggregated over two

seasons (2014/2015 and 2015/2016), yet not outperformed, despite the promising results. In addition, the flow of the model is very important for the prediction process (figure2.10).

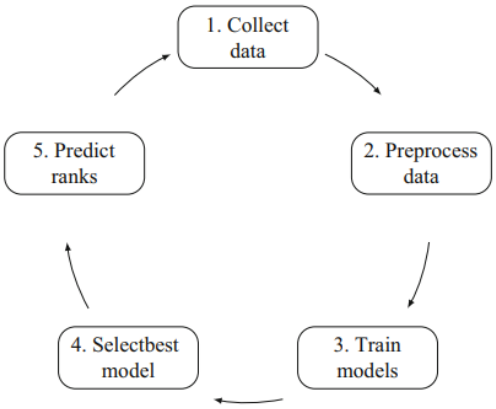


Figure 2.10: Model Flow, **Maanijou and Mirroshandel (2019)**

The methodology followed by researchers is diverse but common attributes were selected to provide a greater means of orientation for the workmanship to ease the reader. The study made by Eryarsoy(2019) on Predicting a Football Game depicts a clear view of a general methodology and would inspire to highlight as below (figure2.11). The same reveals the capturing data, and the preprocessing work to be done including cleansing, missing value imputation, feature engineering, Feature selection, scaling, encoding categorical data, concatenating them and splitting into suitable ratios as training, validation and testing data, Model building with selected algorithms, and finally evaluation to get the best model for the application. The predictive power is assessed with cross-validation.

Alfredo(2019) and Buyrukoglu(2023) expressed that the model evaluations of the football score predictions have been done quite identically for each regression and classification task. In the case of classification, the prediction Accuracy, Precision, Recall and the F1-score were predominant. k -fold Cross validation process has been used in the model validation, the statistical process of evaluation and comparison of the algorithm in the learning process by dividing the data into k- number of folds (where the first fold is reserved for the learning process and the rest for the validation) (figure2.12).

However, Vaidya(2016) claimed that in many cases, it was observed the prediction indexes with respect to the ”Draw”s are severely depleted.

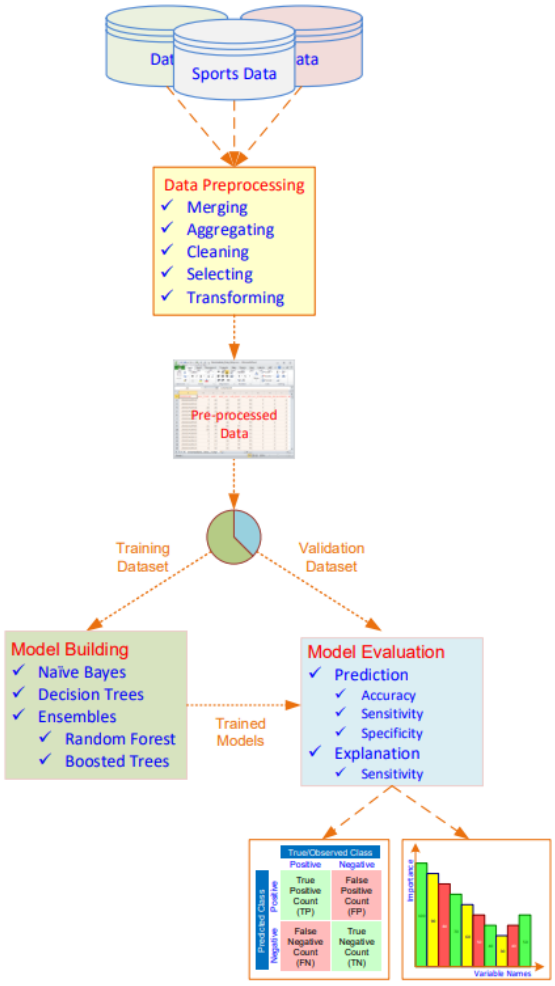


Figure 2.11: General Methodology, **Eryarsoy and Delen(2019)**

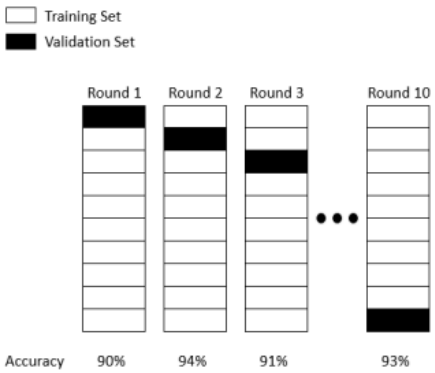


Figure 2.12: Illustration of k-Folds Cross Validation, **Alfredo(2019) and Buyrukoglu(2023)**

Furthermore, in the latest publication, Buyrukoglu(2023) used the ROC curve for the evaluation process as it does not make any impact based on the class distribution and the stacked models being able to secure better results in the study.

2.3.1 Limitations and Future work

A list of features recommended in improving team performances includes ”Increase Ball Possession, Increase pass attempted, Increase pass training, Decrease Fouls, Increase corner attempted, Avoid Offside, Increase shot attempted, Avoid shoot off target, Avoid block, and Increased inside shot attempted, in their study secured over 70 per cent of accuracy and precision (Pugsee and Pattawong 2019).

2.4 Chapter summary

The Literature Review was enveloped with key areas on the empirical evidence from the origination of sports, football, the traditional means of prediction and its replacement with sophisticated ML algorithms, the common algorithms used, the selection of EPL data source as a most-watched game, feature engineering work needed to align the raw data into worthy features, feature selection through sophisticated models alike SelectKBest, up on getting intuition through correlation metric and PCA analysis. Eventually, it was observed that greater utility of commonly used supervised learning algorithms such as Logistic regression, GNB, SVM, k-NN, GBC, ANN, MLP, and Voting Classifiers. Additionally, the Elo rating for the players' ranking and clustering have been used in the prediction process where applicable. finally, a set of matrices including CV scores and ROC-AUC curves were considered in evaluating the models in the referred studies. Observed limitations were marked finally in view of revealing the research gaps.

Chapter 3

UNDERSTANDING THE DATA

3.1 Source

The process of sourcing a suitable dataset was a strenuous effort with the features suite for processing through ML to reveal a robust outcome. EPL, French League 1(France), Spanish La Liga (Spain), Italian Seria A(Italy) and German Bundesliga (Germany) have been the most famous league tournaments in the world where EPL is the most watched premier league in the world with 4.7 billion viewers by the year 2022 (Rudrapal et al. 2020). The dataset was last updated on 28/05/2023 and the data content was on Premier League, Championship, League1, League2 and Conference divisions. The observed key features, Division, Date, Home and away teams, Half and full-time respective goals, match half and full-time results, referee, home and away team shots and shots on target, fouls, red, and yellow cards, betting results, odds and odds offered in 3 betters were significant features. Hence, it was opted the dataset from the EPL from the 1993 to 2022 seasons (https://www.football-data.co.uk/englandm.php), as an open source for this study. It is found that the data content is homogeneous from 2013 to 2022 with 45922 observations and the same was extracted, considering the ease of processing. The sources of betting odds made the feature combination more complex, despite the indirect contribution, hence combining them sensibly, leaving to mark a due contribution. Further, a study made on ”Incorporating domain knowledge in machine learning for soccer outcome prediction” by Berrar et al.(2019) suggests that successful incorporation of the domain knowledge into the ML model would be the key factor to reveal better outcomes. Accordingly, a few more new features were derived through feature engineering and the most influential features over the intuition gained through the correlation matrix to reveal the right subsets of data.

3.2 Ethical and social aspects

3.2.1 Regulation and Governance

The existing data governance regulations do impose personal rights, fairness, and no biases in the data processing whilst preserving human rights as no one is allowed to use someone's bio-data for such a use. In this study, neither personal data nor interviewing was taken and used as open-source data (https://www.football-data.co.uk/englandm.php).

3.2.2 Unintended Consequences

The model uses 10 years of EPL data and the techniques embedded with the model deliver marked results. It cannot be guaranteed the fact that more data will reveal better results. Nevertheless, there could be a great possibility of producing more robust predictions as bigger the data set, where, betting and gambling behaviours could flourish with the model usage.

3.2.3 Ability to reproduce and Open Science

It is indeed necessary to reproduce the match results to meet the objectives of many interested parties rooted in football. The deliberate Jupiter notebook would excel in a comprehensive understanding of the training of the selected ML algorithms. The best model could be used with specific match predictions, which entails the model demonstration, aided by a graphical user interface and reproducibility.

3.3 Data Exploration

The EPL data consists of five divisions of matches (E0, E1, E2, E3 and Ec) listed under Home and Away teams, based on the venue of the match that had been played. It shows 45952 observations with 130 features, 64 integers and 8 object types (figure**??**). For classification cases, the FTR would be the target and the features such as Full Time Home Goal(FTHG), Full Time Away Goal (FTAG), Half Time Home Goal (HTHG), and Half Time Away Goal (HTAG) felt highly correlated with the target FTR, as per the correlation matrix. However, it is open to using the numerical target as FTHG/ HTHG in case one needs to deal with it as a regression process. T

3.3.1 The Main Events of Football Game

Let's identify the important features with the definitions(https://www.football-data.co.uk/englandm.ph

1. Shots: A shot occurs when a player intentionally kicks the ball in the direction of the goalpost of the opposing team. There are numerous distinct types of shots, each struck from a different distance or angle or with a different section of the body.

2. Passes: A pass occurs when a player kicks the ball to a teammate.

3. Crosses: A cross is when a player kicks the ball from the side of the field towards the goal of the opposing team with the objective of passing it to a teammate.

4. Ball Possession: The amount of time a team has the ball during a game is referred to as possession.

5. Penalties and Free Kicks: When a player commits a foul, a free kick is awarded to the field's opposing team. In that situation, the team that concedes the foul may start play from the spot where the foul occurred. If the offence occurs inside the penalty box (the area close to the goal), a penalty is given: the team that admitted liability for the foul has the opportunity to shoot at the goal from close range without any members of the opposing team in the area.

6. Cards: Cards are given out if the referee determines that a foul is significant enough. Smaller fouls result in yellow cards, which have no immediate repercussions. However, a player receives a red card if they receive two yellow cards. Whenever a player receives a red card, they are not allowed to play.

7. Corners: When a side kicks the ball off the pitch behind their goal, the opposing team is given a corner. In this scenario, the ball is positioned on the pitch's corner and is accessible to players alone. The abbreviated format of features could be listed for ease of understanding

Mainly, the features could be divided into 3 categories; Root variables, The variables associated with bookmakers' odds, and Specific Variables, apart from newly derived features through averaging and feature engineering work done to get the precise subsets of data.

3.3.2 16 Root variables

1. FTHG: Full Time Home Goal
2. FTAG: Full Time Away Goal
3. HTHG: Half Time Home Goal
4. HTAG: Half Time Away Goal
5. HS: Home Team Shots
6. AS: Away Team Shots
7. HST: Home Team Shots on Target
8. AST: Away Team Shots on Target
9. HF: Home Team Fouls Committed
10. AF: Away Team Fouls Committed
11. HC: Home Team Corners
12. AC: Away Team Corners
13. HY: Home Team Yellow Cards
14. AY: Away Team Yellow Cards
15. HR: Home Team Red Cards
16. AR: Away Team Red Cards

3.3.3 The variables associated with bookmakers' odds

1. B365H: Bet365 odds favouring the home team
2. B365D: The draw's odds at Bet365
3. B365A: Bet365's odds for the defeating team
4. BWH: Bet and Win's odds for the victory of the home team
5. BWD: Bet and Win's odds for the draw.

6. BWA: Bet and Win odds for the victory of the visiting team
7. IWH: Interwetten's odds for the home team to triumph
8. IWD: Interwetten's odds for the draw.
9. IWA: Interwetten's odds for the victory of the away team
10. LBH: Odds offered by Ladbrokes for Home Team to win
11. LBD: Odds offered by Ladbrokes for the draw
12. LBA: Odds offered by Ladbrokes for Away Team to win
13. PSH: Pinnacle Sports odds for Home Team to win
14. PSD: Pinnacle Sports odds for the draw
15. PSA: Pinnacle Sports odds for Away Team to win
16. WHH: Odds offered by William Hill for Home Team to win
17. WHD: Odds offered by William Hill for the draw
18. WHA: Odds offered by William Hill for Away Team to win
19. VCH: Odds offered by VC Bet for Home Team to win
20. VCD: Odds offered by VC Bet for the draw
21. VCA: Odds offered by VC Bet for Away Team to win

3.3.4 Specific Variables

1. BbMxH: Maximum odds offered by bookmakers for the home team to win.
2. BbAvH: Average odds offered by bookmakers for the home team to win.
3. BbMxD: Maximum odds offered by bookmakers for a draw.
4. BbAvD: Average odds offered by bookmakers for a draw.
5. BbMxA: Maximum odds offered by bookmakers for the away team to win.
6. BbAvA: Average odds offered by bookmakers for the away team to win.

7. BbOU: Over/Under betting market, indicating the total number of goals expected in a match.
8. BbMx: Maximum odds offered for the match (either home win, draw, or away win).
9. BbAv: Average odds offered for the match (either home win, draw, or away win).
10. BbAH: Asian Handicap market, which provides a goal advantage or disadvantage to a team.
11. BbAHh: Handicap value for Asian Handicap, representing the goal advantage or disadvantage.
12. BbMxAHH: Maximum odds offered by bookmakers for Asian Handicap Home Team.
13. BbAvAHH: Average odds offered by bookmakers for Asian Handicap Home Team.
14. BbMxAHA: Maximum odds offered by bookmakers for Asian Handicap Away Team.
15. BbAvAHA: Average odds offered by bookmakers for Asian Handicap Away Team.
16. PSCH: Pinnacle Sports closing odds for the home team to win.
17. PSCD: Pinnacle Sports closing odds for a draw.
18. PSCA: Pinnacle Sports closing odds for the away team to win.

3.3.5 Data Heads

Eventually, the data heads were inspected including date, away team or home team, primary features, betting odds and specific features. The features were inspected for missing values and outliers. The correlation matrix was used to get an overview of the feature independence needed toward the target match result in view of excluding them from the model. Domain knowledge is highly important and getting acquainted with the feature is wiser as insight into would make better end results (figure3.1).

```
1 fb_df.head()
```

	Div	Date	HomeTeam	AwayTeam	FTHG	FTAG	FTR	HTHG	HTAG	HTR	...	AvgC<2.5
0	E0	17/08/13	Arsenal	Aston Villa	1.0	3.0	A	1.0	1.0	D	...	NaN
1	E0	17/08/13	Liverpool	Stoke	1.0	0.0	H	1.0	0.0	H	...	NaN
2	E0	17/08/13	Norwich	Everton	2.0	2.0	D	0.0	0.0	D	...	NaN
3	E0	17/08/13	Sunderland	Fulham	0.0	1.0	A	0.0	0.0	D	...	NaN
4	E0	17/08/13	Swansea	Man United	1.0	4.0	A	0.0	2.0	A	...	NaN

5 rows × 130 columns

Figure 3.1: Feature heads,

3.3.6 Data Count

The data count would be more logical to reveal the number of observations (45952), any missing data, and the types of data (130 floats and 8 objects) (figure3.2). The features that do not have the above number of observations would be the features with missing data.

```
1 fbdata.count()
```

```
Date         45952
HomeTeam     45952
AwayTeam     45952
FTHG         45952
FTAG         45952
             ...
PCAHA         9841
MaxCAHH       9856
MaxCAHA       9856
AvgCAHH       9856
AvgCAHA       9856
Length: 138, dtype: int64
```

```
1 # Identify the data types
2 fbdata.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 45952 entries, E0 to EC
Columns: 138 entries, Date to AvgCAHA
dtypes: float64(130), object(8)
memory usage: 48.7+ MB
```

Figure 3.2: Data Count,

3.3.7 Summary Statistics

The summary statistic would entail great intuition on the range of the features on mean, standard deviation, and quantile positions of the data, (figure3.3).

	FTHG	FTAG	HTHG	HTAG	HS	AS	HST	AST	\
count	45952.0	45952.0	45951.0	45951.0	42309.0	42309.0	42309.0	42309.0	
mean	1.4	1.2	0.6	0.5	12.3	10.1	5.3	4.3	
std	1.2	1.1	0.8	0.7	4.8	4.3	2.9	2.5	
min	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
25%	1.0	0.0	0.0	0.0	9.0	7.0	3.0	2.0	
50%	1.0	1.0	0.0	0.0	12.0	10.0	5.0	4.0	
75%	2.0	2.0	1.0	1.0	15.0	13.0	7.0	6.0	
max	9.0	9.0	7.0	5.0	43.0	35.0	24.0	20.0	

	HF	AF	...	AvgC<2.5	AHCh	B365CAHH	B365CAHA	PCAHH	\
count	42307.0	42307.0	...	9856.0	9856.0	9839.0	9839.0	9841.0	
mean	11.1	11.6	...	1.9	-0.2	1.9	1.9	1.9	
std	3.7	3.8	...	0.3	0.6	0.1	0.1	0.1	
min	0.0	0.0	...	1.4	-3.8	1.6	1.3	1.6	
25%	9.0	9.0	...	1.7	-0.5	1.8	1.8	1.9	
50%	11.0	11.0	...	1.8	-0.2	1.9	1.9	1.9	
75%	13.0	14.0	...	2.0	0.2	2.0	2.0	2.0	
max	33.0	77.0	...	5.8	3.0	3.6	2.4	3.9	

	PCAHA	MaxCAHH	MaxCAHA	AvgCAHH	AvgCAHA
count	9841.0	9856.0	9856.0	9856.0	9856.0
mean	2.0	2.0	2.0	1.9	1.9
std	0.1	0.1	0.1	0.1	0.1
min	1.3	1.7	1.3	1.7	1.3
25%	1.9	1.9	1.9	1.8	1.8
50%	1.9	2.0	2.0	1.9	1.9
75%	2.0	2.1	2.1	2.0	2.0
max	2.4	4.0	2.5	3.7	2.2

[8 rows x 130 columns]

Figure 3.3: summary statistics,

3.3.8 Performance Analysis

The Team performance is very important in predictions, where respective team performance against feature heads are displayed (figure3.4). The figure indicates the average goals, shots and shots on target against teams

The team comparative performance would be embedded with the competing power among the teams (figure3.5).

The win-loss analysis would ultimately mark the story talking about, where 19936/45952 home wins, 13892/45952 Away wins and 12124/45952 Draws are secured (figure3.6).

The home vs. Away Team Shots were inspected as found contributory (figure3.7).

3.3.9 Correlation Analysis

The correlation analysis would give the intuition on selecting the essential independent feature, based on the scale of the relationships (figure3.8).

	HomeTeam	FTHG	FTAG	HTHG	HTAG	HS	\
0	AFC Telford United	1.043478	1.130435	0.347826	0.521739	8.565217	
1	AFC Wimbledon	1.329114	1.265823	0.582278	0.556962	11.259494	
2	Accrington	1.534483	1.330049	0.662562	0.544335	12.487685	
3	Aldershot	1.395062	1.232099	0.632099	0.570370	9.071713	
4	Alfreton Town	1.717391	1.739130	0.869565	0.750000	11.054348	
..	...	...	...	...	...	...	
154	Wolves	1.379581	1.162304	0.638743	0.518325	12.612565	
155	Wrexham	1.586207	0.903941	0.753695	0.413793	9.416667	
156	Wycombe	1.344743	1.095355	0.633252	0.444988	10.850856	
157	Yeovil	1.144963	1.184275	0.538084	0.486486	9.403727	
158	York	1.334459	1.097973	0.611486	0.449324	11.976000	

	AS	HST	AST	HF	AS	HF	\
0	10.347826	4.913043	5.565217	11.913043	10.347826	11.913043	
1	9.962025	4.481013	4.075949	11.734177	9.962025	11.734177	
2	10.167488	5.371921	4.386700	10.906404	10.167488	10.906404	
3	6.585657	4.290837	3.159363	10.462151	6.585657	10.462151	
4	11.217391	5.489130	6.032609	11.097826	11.217391	11.097826	
..	...	...	...	...	...	...	
154	10.900524	5.277487	4.730366	11.295812	10.900524	11.295812	
155	6.265873	4.178571	2.821429	11.190476	6.265873	11.190476	
156	8.540342	4.400978	3.481663	12.078240	8.540342	12.078240	
157	8.863354	4.133540	3.835404	11.593168	8.863354	11.593168	
158	8.852000	5.820000	4.220000	10.076000	8.852000	10.076000	

	AF	HC	AC	HY	AY	HR	AR
0	11.217391	4.956522	4.869565	2.086957	1.826087	0.086957	0.217391
1	11.000000	5.620253	4.803797	1.354430	1.734177	0.041139	0.123418
2	12.421182	5.512315	5.128079	1.413793	1.467980	0.110837	0.098522
3	10.697211	5.601594	4.784861	1.392593	1.735802	0.074074	0.101235
4	10.815217	5.673913	5.478261	1.402174	1.500000	0.130435	0.108696
..	...	...	...	...	...	...	...
154	10.900524	5.952880	4.643979	1.353403	1.845550	0.047120	0.091623
155	11.940476	5.789683	4.686508	1.401478	1.709360	0.088670	0.100985
156	12.029340	5.836186	4.728606	1.359413	1.809291	0.063570	0.105134
157	11.527950	5.257764	5.270186	1.584767	1.776413	0.103194	0.122850
158	11.668000	6.036000	4.496000	1.236486	1.706081	0.060811	0.077703

[159 rows x 19 columns]

Figure 3.4: Performance Analysis of Teams,

	Team	AvgHomeGoals	AvgHomeShots	AvgHomeShotsOnTarget	\
0	AFC Telford United	1.043478	8.565217	4.913043	
1	AFC Wimbledon	1.329114	11.259494	4.481013	
2	Accrington	1.534483	12.487685	5.371921	
3	Aldershot	1.395062	9.071713	4.290837	
4	Alfreton Town	1.717391	11.054348	5.489130	
..	...	...	...	...	
154	Wolves	1.379581	12.612565	5.277487	
155	Wrexham	1.586207	9.416667	4.178571	
156	Wycombe	1.344743	10.850856	4.400978	
157	Yeovil	1.144963	9.403727	4.133540	
158	York	1.334459	11.976000	5.820000	

	AvgAwayGoals	AvgAwayShots	AvgAwayShotsOnTarget
0	0.913043	8.043478	3.347826
1	1.073016	9.625397	3.926984
2	1.073710	10.496314	4.255528
3	1.176904	9.466135	4.677291
4	0.989130	7.869565	3.760870
..	...	...	...
154	1.152231	10.425197	4.301837
155	1.216749	9.472222	4.722222
156	1.208845	10.611794	4.343980
157	1.068796	9.338509	4.288820
158	1.209459	8.908000	4.192000

[159 rows x 7 columns]

Figure 3.5: Home vs Away Performance,

2.4. Win loss Analysis

```
1 outcome_frequency = fbdata['FTR'].value_counts()
2 print(outcome_frequency)
```

```
H    19936
A    13892
D    12124
Name: FTR, dtype: int64
```

Figure 3.6: Home win, Draw and Away win analysis,

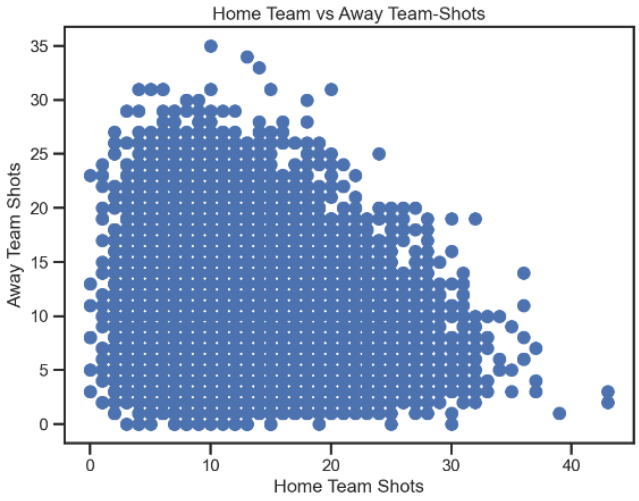


Figure 3.7: Home vs Away Team Shots,

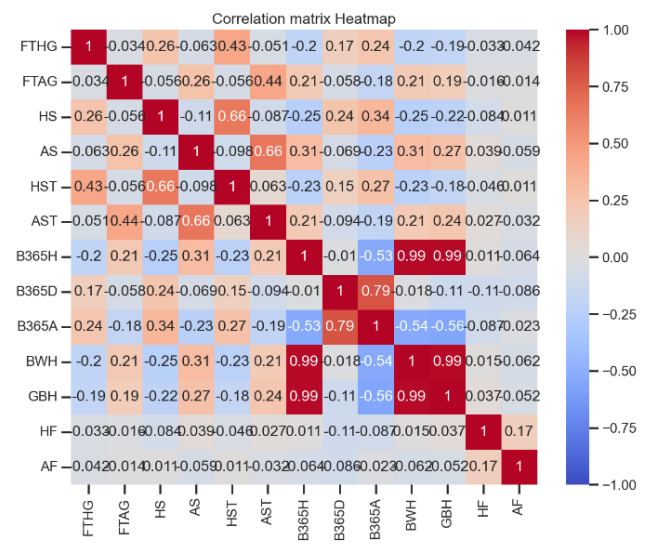


Figure 3.8: Correlation Matrix,

3.3.10 Other Analysis

Other analyses were done to be acquainted with the importance of Referee and betting odds (figure3.9).

2.7. Referee. Analysis

```
Referee_statistics = fbdata.groupby('Referee').mean()[['HF', 'AF', 'HY', 'AY', 'FTHG', 'FTAG']]
print(Referee_statistics)
```

```
                HF         AF        HY        AY      FTHG      FTAG
Referee
A Ajibola      NaN        NaN  1.363636  1.954545  1.545455  1.090909
A Allison    10.500000  11.500000  1.000000  2.000000  0.500000  0.000000
A Backhouse  10.932961  11.145251  1.427136  1.567839  1.482412  1.105528
A Bannister    NaN        NaN  1.400000  1.600000  1.333333  1.266667
A Bates      10.381818  11.650000  0.981818  1.609091  1.386364  0.990909
...                ...        ...       ...       ...       ...       ...
W Barrett    11.875000  13.625000  1.250000  1.250000  1.875000  0.750000
W Cartmel      NaN        NaN  1.000000  1.312500  1.687500  0.937500
W Finnie     10.902913  11.485437  1.392000  1.696000  1.416000  1.208000
k Stroud     12.000000  19.000000  0.000000  2.000000  0.000000  0.000000
l Mason      11.000000  10.000000  4.000000  1.000000  1.000000  1.000000

[480 rows x 6 columns]
```

2.8. Odd Analysis

```
odds_columns = ['B365H', 'BWH', 'GBH']
odds_comparision = fbdata[['FTHG', 'FTAG'] + odds_columns]
print(odds_comparision)
```

```
      FTHG  FTAG  B365H   BWH   GBH
Div
E0     2.0   2.0   2.30  2.10  2.25
E0     0.0   2.0   5.00  4.35  4.25
E0     0.0   0.0   2.37  2.30  2.30
E0     0.0   0.0   1.72  1.65  1.73
E0     0.0   0.0   2.87  2.90  2.70
..     ...   ...    ...   ...   ...
EC     1.0   1.0   1.25  1.21   NaN
EC     1.0   1.0   2.70  2.55   NaN
EC     0.0   1.0   3.80  3.50   NaN
EC     2.0   1.0   1.50  1.44   NaN
EC     1.0   1.0   7.50  5.50   NaN

[45952 rows x 5 columns]
```

Figure 3.9: Other Analysis,

However, it is observed with missing values and outliers and is to be handled during the data proceeding (figure3.10).

3.1. Addressing missing values, Outliers and the Noises

```
fb_df.isnull().sum()
```

```
Div            14
Date           14
HomeTeam       14
AwayTeam       14
FTHG           14
              ...
PCAHA       15557
MaxCAHH     15542
MaxCAHA     15542
AvgCAHH     15542
AvgCAHA     15542
Length: 130, dtype: int64
```

Figure 3.10: Missing Value Observed,

3.4 Chapter Summary

The understanding of the data is paramount prior to undertaking preprocessing work to make the task ahead to be accommodated meaningfully. Accordingly, a logical process was followed during the data understanding process. The chapter started with understanding the ethical and social considerations needed and found it as open source data of most watched EPL matches with categories of features such as root, betting odds, and specific variables. The features included 130 floats and 8 object types where the selected categorical targets Full Time Results(FTR), FTHG, FTAG, HST, AST, HomeTeam, Away Team, HC, and AC" with 45952 rows of observations. The feature distribution, summary statistics, winning status, team performances, and visualization of data with correlations were highlighted and the missing values need to be addressed prior to moving ahead.

Chapter 4

DATA PREPROCESSING

4.1 Missing Value Imputation, Handling Outliers and the Noises

Data preprocessing is an essentially important step in arranging the data and alignment to proceed with the model training process smoothly (Alexandropoulos, Sotiris B. Kotsiantis, and Vrahatis 2019). Alexandropoulos(2019) emphasises that the missing value imputation takes precedence, to effect further processing. The numerical and categorical features are in the data set and they are to be handled separately over the difference in nature of data. Let's take the categorical data first as only 5 essential features to be addressed. The mode value seemed the best option for imputing categorical features. Hence, the 'HomeTeam', 'AwayTeam', 'FTR', 'HTR', and 'Referee' were imputed with the mode values of the respective feature. Handling the numerical features was taken eventually and the removal of the outliers and the noises were considered, to avoid unnecessary fatigue on missing value imputation. The same was addressed in three steps as Root features were taken as the first step, considering the threshold weights applied for removing the skirting, since losing the margin of observations is greatly important, based on the total number of observations. However, it was revealed there was no appropriate logic for the task and even the model accuracies dropped during the trial process. The bedding odds were treated eventually, with different weights upon a trial and error process. Subsequently, the missing value imputation of the numerical data was taken care of, based on the mean value of the respective numerical data whilst the 'Date' was filled with the "afill"/ "bfill" functions as found appropriate.

4.2 Feature Engineering

Verdonck et al. (2021) define it as ”clever engineering of data exploits, intrinsic biases of the ML technique to greater benefit in terms of accuracy and interpretability”.

4.2.1 Shrinking Betting Odds to Eliminate the Feature Complexity

The features of the data set were dominated by the betting odds elevating the complexity of the commercial objectives. It is perceived that averaging them suitably would retain the contribution rather than dropping them. Accordingly, a few different categories of betting odds were averaged as representing Home, draw, away win, Maximum odds, and odds over 2.5 goals, to retain the context of the homogeneous bunch of features (figureB.2).

4.2.2 Deriving the Other Features in Align with predictions

The Goal Gap The goal gap was perceived as another aspect to be considered over the empirical evidence to be derived (figureB.3). However, the feature was not selected for training models as found propensity of overfitting, as evidenced by the correlation of features and results.

The Winning Streak The winning streak could be another important attribute to be considered. Away goal conceded average and averaged goals scored were popular among the historical evidence. The ’Date’ feature is essential but the form to be converted to ordinal to make sense by making it ordinal to match with floats. Hence the date time function in Python was used to form the feature (figureB.4).

Weighted Age However, it is an obvious fact that the recent events are to be outweighed to elevate the contribution to gain the currency effect. Accordingly, the Weighted age was derived and subsequently, the ’Date’ feature was dropped.

Points Score The points scored by the home and the away team based on the goals scored, could be another important Key Performance Indicator (KPI) and hence the same was introduced. Calculating the probabilities of winning, through the Poisson distribution was considered later in the process, in view of elevating the intuition of the reader.

Form of a Team The form of a team is essentially important in winning and losing trends and one unique feature, the 'Form Indicator' is derived, additionally (figureB.5). The point difference in the last 5 matches was considered for the calculation to simulate the weight and the form, by mapping categories (figure4.1).

Age	HomeTeam	AwayTeam	FormIndicator
0	Torquay	Wrexham	6
0	Derby	Morecambe	-7
0	Exeter	Charlton	-7
0	Fleetwood Town	Burton	-7
0	Forest Green	Peterboro	-7
...	...	...	...
10	Crawley Town	Port Vale	-7
10	Crewe	Stevenage	-7
10	Gillingham	Notts County	-7
10	Oldham	Carlisle	-7
10	Arsenal	Aston Villa	7

398 rows x 4 columns]

Figure 4.1: Team Form,

4.2.3 ELO Rating

Betting markets in sports are increasingly competitive over the interest shown by the investors and Elo rating is used to derive covariates that are used in the ordered logit regression model. Arpad Elo designed a rating for calculating the relative skill levels of individual players with a higher rating for the stronger player (Hvattum and Arntzen 2010). It started with Chess but it has been propagated down to other games including football as widely adapted since it provides productive and straightforward means to foresee and track the relational skill level of individual players or competitors over time (Hvattum and Arntzen 2010; Kovalchik 2020). The process of work takes a few steps as below.

During the initialisation, a predefined figure is primarily assigned for each player, in most cases based on general awareness, which could vary on different applications. As the Outcome Prediction, the probability of each player winning the match based on the set value is calculated using the logistic function, before the match. The actual outcome of the match is used for adjusting the rating of the individual player, based on the match results expectation and the outcome. In case the expectation was to lose and if wins the match, the margin of rating increase would be high and vice versa, otherwise. The rating would be reduced in case of a match based on the above criterion.

The sensitivity of the rating is adjusted to each match outcome where a new player may get a higher K factor as yet establishing whilst it would be otherwise for a veteran. Hence a gradual deviation could be expected against an experienced player. The skill level of a player does not change drastically, even though observed with losing series. Hence a threshold level is established to preserve the player as a Rating Floor. The Elo rating would be periodically updated regularly to preserve the currency. A study on "A Rating System For Gaelic Football Teams: Factors That Influence Success" in Ireland by Mangan and Collins(2016), was able to secure 72.90 per cent of accuracy. However, the accuracy gained in this study terminated using the feature, considering the elevated results marked.

4.3 Feature Selection

Alfredo(2019) states the importance of feature selection as it has been a unique segment in the machine learning processes to identify the most influential and contributory features towards the target realization and has been evident in almost all the research. It is better to be acquainted with the correlation matrix, prior to using the feature selection to get better intuition.

4.3.1 Correlation Matrix

Pugsee(2019), states that the correlation matrix is a better tool for getting sound intuition among features towards the target. However, the categorical target is left alone and foresees the relationship among others as a glimpse of understanding, keeping the FTHG and the FTAG on top as they are highly correlated with the target. The figures in the correlation matrix (+1) to (-1) range where the (+1) indicates ultimate collinearity the (0) indicates no collinearity and the (-1) indicates the ultimate negative or inverse collinearity. In case one checks these figures with FTHG or FTAG, the numerical representation would give better intuition about the contribution of the respective feature towards the target (FTR) outcome. Other than root features, some other derived attributes are surfaced, showing their productivity, in terms of production. The Heatmap adds value to the correlation identification as given below, being a visual representation of the Correlation Matrix, using colours to represent the relationship. In the case here, it visualizes the high collinearity with dark colours towards the red and otherwise the blue. All the ones away from the diagonal line are duplicated features and are to

be avoided in future selections(figure4.2).

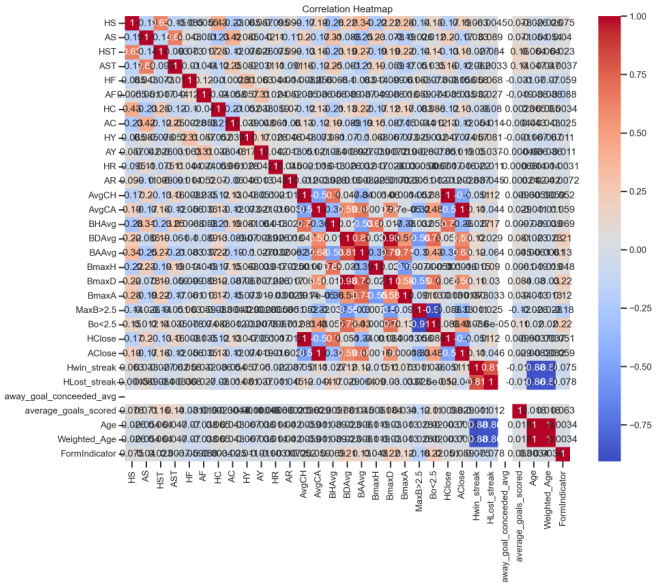


Figure 4.2: Heatmap,

4.3.2 SelectKBest

Upon intuition, the ”SelectKBest” tool was used based on the empirical evidence stated and for the task more productively, as found suitable for selecting features towards the prediction of a categorical target. Accordingly, the tool primarily selected the below features (figure4.3).

```
Top 20 Selected Feature Names:
Index(['FTHG', 'FTAG', 'HTHG', 'HTAG', 'HS', 'AS', 'HST', 'AST', 'HY', 'HR',
       'AR', 'AvgCH', 'AvgCA', 'BHAvg', 'BDAvg', 'BAAvg', 'BmaxH', 'BmaxD',
       'BmaxA', 'HClose', 'AClose', 'goal_gap', 'Hwin_streak', 'HLost_streak',
       'average_goals_scored'],
      dtype='object')
```

Figure 4.3: Top 20 Selected Features,

The below graph is evident for witnessing the behaviour of the selected features after removing highly correlated features. There was no sound logic to remove the outliers and the noises surfaced, over the nature of the features and even depleted the results in trials and hence left alone (figure4.4).

However, the correlation matrix suggests the level of close collinearity of certain features (’HTR’, ’FTAG’, ’FTHG’, ’HTAG’, ’HTHG’) would be overfitting over the homoscedasticity in model fitting, and hence the same to be avoided from the selected list (figure4.5).

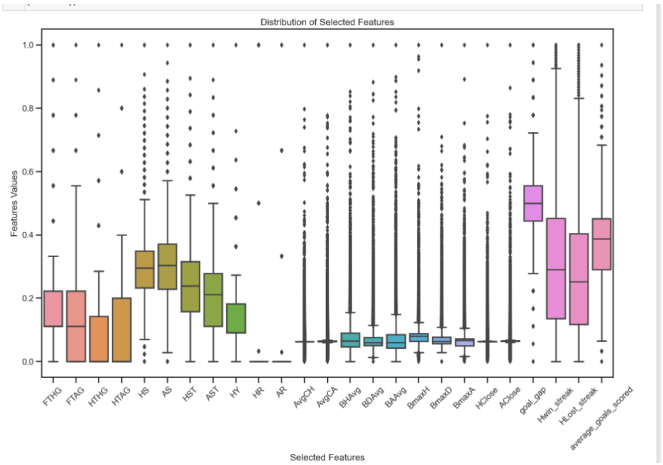


Figure 4.4: Selected Features,

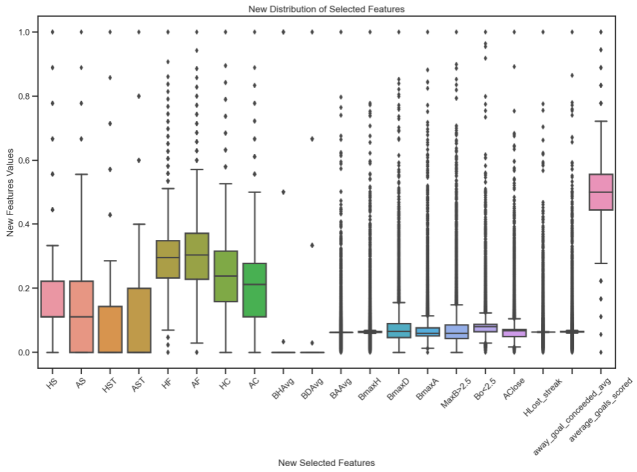


Figure 4.5: Finalized Selected Features,

4.3.3 The Principle Component Analysis(PCA)

Principle Component Analysis (PCA) transforms the original features into a new set of features called the principle components, which are linear components of the original components. In view of perceiving the number of actual contributory lower dimensional features, PCA configures a final list of features out of already selected, reducing the dimensionality (figureB.6). The explained variance could be tabulated as below, (figure**??**), indicating a lesser number of contributory features.

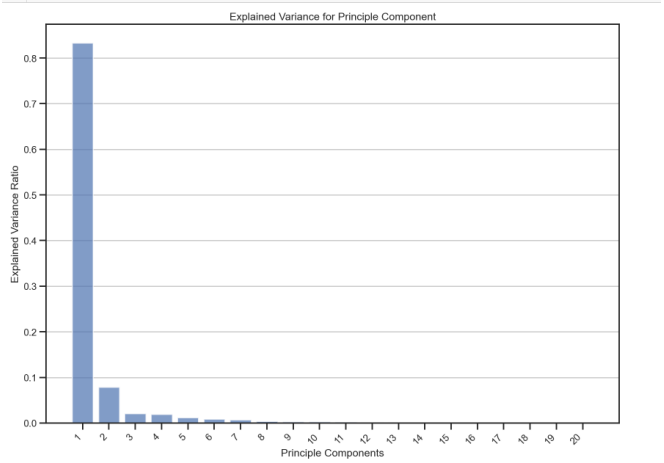


Figure 4.6: Explained Variance for Principle Components,

However, the analysis revealed that there would only be 17-20 features a making difference in the final results and with a safer margin, the selected feature list is restricted (figure4.7).

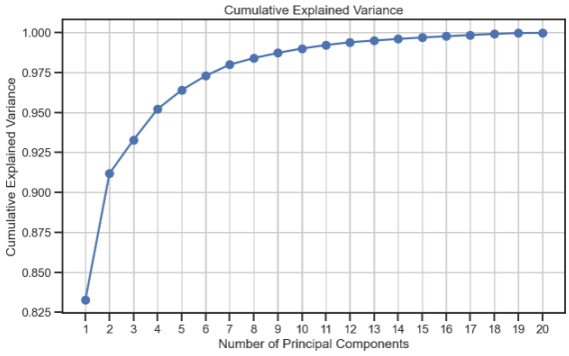


Figure 4.7: Cumulative Explained Variance,

The contribution of the original components towards the principle components can be demonstrated below. When it is 1, it is fully correlated and otherwise it is fully

uncorrelated where the rest is proportionally related(figure4.8).

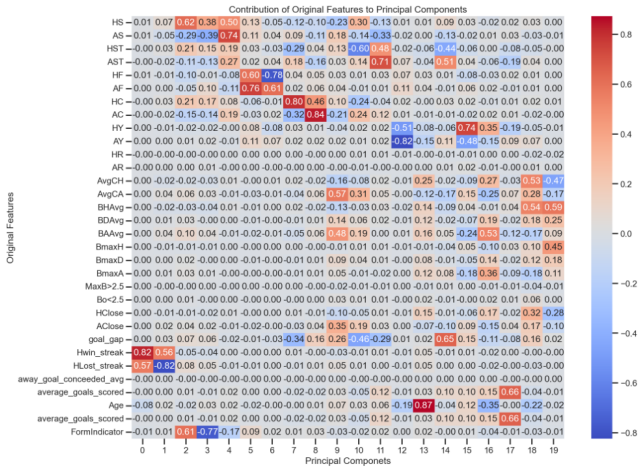


Figure 4.8: Contribution of Orignal Features to Principle Components,

4.4 Establishing data frames

The selected data cannot be fed for any algorithms as it is and to be further aligned to reveal the intended outcome productively, in a manner these algorithms reveal as mainly two categories of data exist, numerical and categorical, which demand different operations. That includes further imputing for any missing values if any, scaling ordinal features, encoding categorical features, and concatenating them (figureB.7).

As per the empirical evidence stated in the literature review, the process demands the training of the model with training data and the resultant is validated against unseen validation data for robustness, most appropriately. Yet it demands the evaluation of the selected best model against a sample of unseen testing data, in selecting the best model out of the individual best model, prior to the application of the selected final model. Hence data is to be split into three portions as below (figureB.8).

4.5 Clustering

4.5.1 Overview of Clustering

Clustering is a division of ML that comes under unsupervised segment use in grouping the data points based on a given intrinsic characteristics, patterns, and similarities, most

importantly without any predefined labels or categories (A. K. Jain 2010). The main objective of the grouping is to discover the inherent structure within the data, paving the way for identifying the inherent structures embedded within the data to use for meaningful groupings. The structures are defined in a way that maximizes the intra-cluster similarity and minimizes inter-cluster similarity, where the data points within the same cluster are more similar to each other compared to those in different clusters.

4.5.2 The Importance of Clustering

A search via Google Scholar found 1960 entries of clustering by 2007 and the vast literature speaks to the importance of clustering, (A. K. Jain 2010). The Importance of clustering in football match results prediction is the ability to provide inherent patterns. However, it contributes to feature engineering and elevating the quality of the overall prediction process. It helps in feature identification by grouping similar matches, based on features and segmenting teams with common strategies or weaknesses that lead to the specific outcomes of the match. The clustering results could be used for the ensemble models as inputs to reveal more robust prediction

4.5.3 The methods of clustering

Various clustering algorithms are evident through each approach and assumption as below.

K-Means Clustering K Means clustering shows a great history of nearly 70 years as first published in 1955, yet extensively in use with the optimization of objective function that improves the quality of the partitions iteratively (A. K. Jain 2010; Oyelade et al. 2019). This is a partitioning-based algorithm that aims to divide the data into K clusters, where, K is a user-defined parameter that iteratively assigns the data points to the clusters, based on their proximity to the cluster centroids and updates the centroids accordingly. The application of the football data here would reveal the course (figure4.9).

Hierarchical Clustering This by its name creates a tree structure of clusters (dendrogram), which commences with each data point as its own cluster and then merges clusters based on their similarity, creating a hierarchy of clusters (Oyelade et al. 2019).

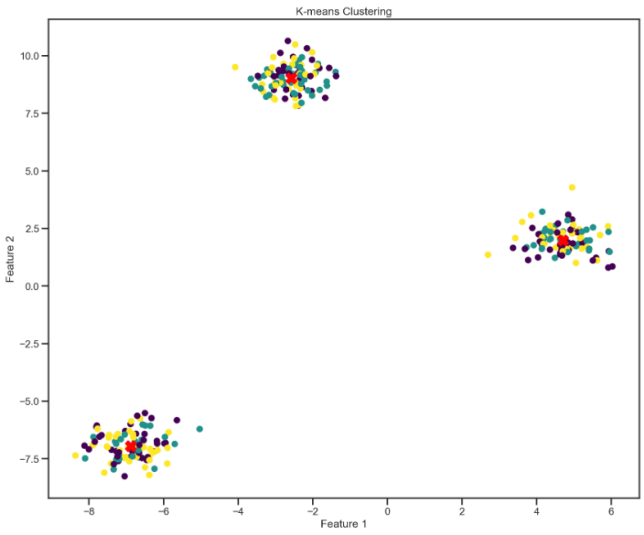


Figure 4.9: K Means Clustering,

The application of the football data here would reveal the tree-based structure of clustering (figure4.10). Further, it is finding other clustering methods such as Density-Based Spatial Clustering of Application with Noise (DBSCAN) which identifies dense regions that handle arbitrary shapes of clusters, Gaussian Mixture Model (GMM), a probabilistic models that generate data points from a mixture of several Gaussian distributions to frame the clusters, are evident.

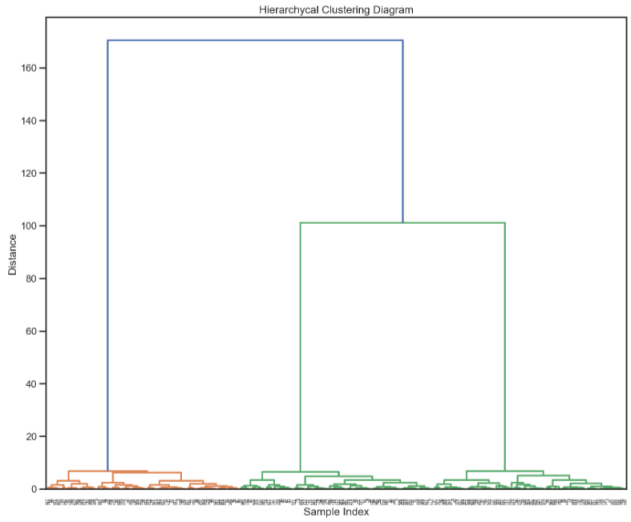


Figure 4.10: Hierachical Clustering,

4.5.4 Cluster Applications

Clusters are applied in many fields such as segmenting target markets in education, business, biology, image segmentation in computer vision, document clustering in natural language processing, Detection of anomalies in fraud detection, grouping genes in bio-informatics, identifying community structures in social networks (Ezugwu et al. 2022). So it has become a powerful tool for uncovering insights from data and organizing complex datasets into more understandable and manageable groups. Eventually, the use of classification works in sports activities has become significant in the past decade as AI revealed. The EPL match results even grouped as below with Home Win (H), Draw (D) and Away Win (A), classes easily, however, the verification process with actual classes was not found to improperly match with win categories. Hence, the application of clustering for this project was not encouraged.

(figure4.11).

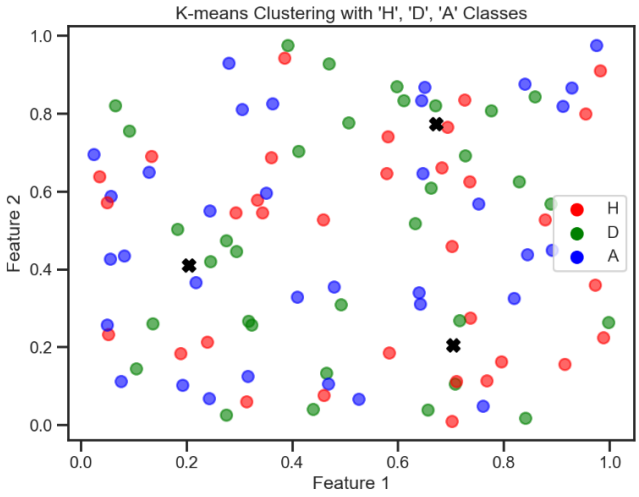


Figure 4.11: K-Means Clustering with Match Results,

4.6 Re-sampling

The trial runs of training models revealed low indices of match draws and indicated a probable class imbalance. Lichtenwalter(2010) stated that class imbalance is very important in revealing better results. Further, it is a rudimentary machine learning technique in manipulating the training data to overcome the class imbalance naturally observed.

4.6.1 Over/ Under/ Combined Sampling

Oversampling elevates the representation of the minority class by adding more synthetic instances. Synthetic Minority Over-sampling (SMOTE) and Adaptive Synthetic Sampling (ADASYN) would be better applications. In contrast, the under-sampling reduces the majority class instances to gain the desired model outcome. Random Undersampling, Tomek Link, and Edited Nearest Neighbours (ENN) are sound examples (Dodangeh et al. 2020). The combination of over and under-sampling has also been applied in some instances.

4.6.2 SMOTE Sampling

SMOTE, SMOTE (ENN), and ADASYN were said to address the class imbalance cases in the use of these algorithms effectively as they identify significantly low samples well, (Z. Wang et al. 2018; Dang et al. 2021). However, the SMOTE was considered for this application as showing great improvement in classification accuracy (by 23 per cent) in an application (Pelayo and Dick 2007) and primarily, it identifies the minority class and generates synthetic instances with the aid of k number of nearest neighbours using Euclidean distance with data points by interpolating selected instances. The process is repeated to realize the desired level of samples. The model training input was settled for 10912 equivalent samples from match result outcome as below, (figureB.9).

```
D    10912
H    10912
A    10912
Name: FTR, dtype: int64
```

Figure 4.12: Input Data after Resampled with SMOTE,

4.7 Chapter Summary

The chapter on data Pre-processing was started by imputing missing values, shrinking betting odds to avoid feature complexity, and introducing a few new features such as form indicator, winning streak, age, weighted Age, goals conceded, and the correlation matrix prior to the application of the SelectKBest in finding most influential 20 features aided with the prediction through PCA application, applying K-Means and hierarchical clustering. Finally, a new data frame was established for training the model with selected 20 features in the next chapter, followed by Synthetic Minority Oversampling(SMOTE).

Chapter 5

METHODOLOGY AND MODEL DESIGN

The literature captured insight into standard methodologies. Accordingly, it is wished to emphasise the flow of work extended in the primary model design phase to get acquainted with the overview needed for the reader. On the selection of the most watched EPL data, a rigorous process of work was carried out to bring them down to a usable effective input through missing value imputation, feature engineering, feature selection, clustering, re-sampling, encoding categorical data, scaling, concatenating and finally splitting the data to be used as training, validation and the testing demanded. The empirical evidence in the literature review stated that the splitting of the data as training, validation and testing in different proportionate and for the case. Accordingly, 70 per cent training, 20 per cent validation and 10 per cent testing data were demarcated over the intuition and the optimized model scores gained through iterative trials. The model training, validation and iterative improvement were carried out, in order to reveal the best model prior to the model application and the constraints were left as proposals for future research, upon serving the academic objectives (figure5.1).

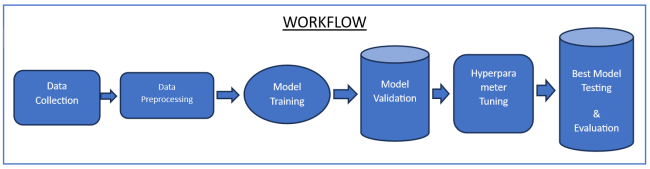


Figure 5.1: Project Workflow,

Accordingly, the model is designed, giving presidency for training five ML models with training data (70 per cent); RFC, k-NN, MLP, GBC and VE, on preprocessing to meet the project objectives upon validation and evaluation (using validation data) through indicated matrixes(figure5.2).

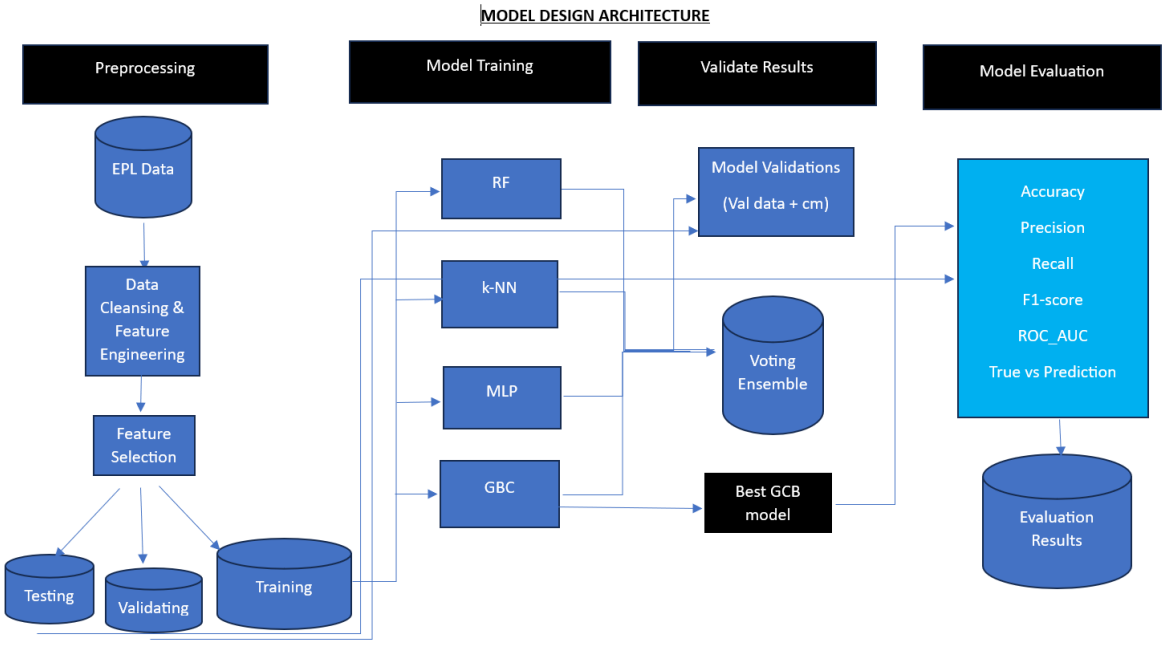


Figure 5.2: Model Design,

5.1 Rationale behind model selection

5.1.1 Random Forest Classifier

Working principle of algorithm RFC is an ensemble classification model embedded with multiple decision trees that have been effectively used in the prediction of football match results (by adequately mapping complex nonlinear decision boundaries), by application of recursive partitioning (Baboota and Harleen Kaur 2019; Wu et al. 2019). The exact position of the cut point and selection of the splitting variable depends on the distribution of observation in the learning sample. This is an unstable algorithm over the course of selecting the first cut point or the first splitting variable would be affected by minute changes in the learning data, changing the structure of the trees. However, the model overcomes the issue by combining a set of highly diverse trees with the averaged results of each tree, since, similar trees will affect the accuracy and hence the diversity demanded (Wu et al. 2019). The model achieves a higher classification accuracy through a randomized training data set and randomized input variable set that reveals multiple new training data subsets through arbitrary extractions and substitutions. The classification results would be voted on all trees, once they are established (figure5.3).

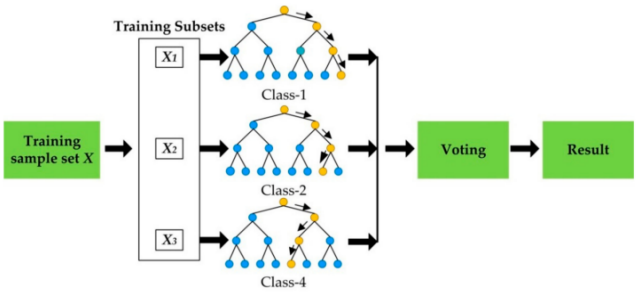


Figure 5.3: Mechanism of Producing Random Forest results, **Wu et al. (2019)**

Further, the model uses a subset of training data with replacement (called bootstrapping) to build individual decision trees and splits at each node whilst avoiding overfitting with the randomization of impregnated subsets. It realizes a larger number of individually trained trees over random subsets. Eventually, it leads to voting prediction through the combination of the outcome of all trees by majority voting, overcoming the issues due to high variance. The higher performance is based on its robustness in overfitting by averaging predictions of individual trees and the embedded

feature importance, scaled through the extent of contribution towards the model outcome. The higher data space with greater dimensions and tolerance to outliers (with the stated advantages) outweigh the drawbacks in the capacity to interpretations of the outcomes, computational complexity and sensitivity to correlated features (Baboota and Harleen Kaur 2019; Wu et al. 2019).

5.1.2 k-NN Classifier

The study made on comparative analysis of the kNN technique emphasised that the model is a very simple one for getting familiar with and using it (Igiri et al. 2015). It functions under the premise that objects or data points with comparable characteristics likely belong to the same class. The idea behind kNN is simply to categorise a new instance, it scans the feature space for the k nearest neighbours and assigns the class label depending on the dominant class among its neighbours. As it is a non-parametric method, it makes no assumptions about the underlying distribution of the data and hence it is applied to a variety of dataset types, regardless of how distributed or complex they are. The flexibility shown in Data Handling is not restricted to binary classification problems and can handle data with any number of classes, hence would be beneficial in this prediction with triple outcomes. The algorithm also handles attributes that are category and numerical. It is a useful tool in many sectors due to its capacity to handle various data kinds. The model does not involve a distinct training process, in contrast to other ML algorithms that do. Instead, the categorization stage makes use of the complete dataset. When working with dynamic or streaming data, this might be helpful because it allows the classifier to swiftly adjust to changes without having to be retrained.

In addition, the k-NN classifier is capable of handling scenarios when the decision boundaries between classes are asymmetrical or irregular. It is especially helpful in circumstances where the underlying linkages are complex or difficult to separate since it may capture complex decision boundaries by taking the local structure of the data into account. The kNN model handles noisy data or outliers relatively well as it can reduce the impact of a single noisy point because it considers the votes of numerous neighbours. Hence, it would be a useful option when working with datasets that could include inconsistent or false samples. It is frequently employed to make product or service recommendations to users, based on how they compare to other users. The model can generate precise recommendations by taking into account the interests or

characteristics of comparable users. The ”lazy learning” paradigm, in which the classifier delays the generalisation phase until a new instance needs to be categorised, is represented by the algorithm. Over the adaptability, kNN is able to effectively handle circumstances involving incremental learning.

Working principle of kNN algorithm

The classification is done by the majority of votes based on the k-value. The neighbours are taken from the set of examples and termed as the training process, even though there is no explicit training to be mentioned. The basic algorithm consists of two steps; finding the k-training example that is closest to the unseen examples. Eventually, it takes the most commonly occurring classification for these k examples, pertaining (figure5.4).

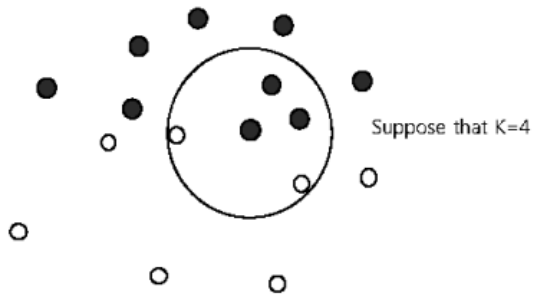


Figure 5.4: k-NN Search Problem for k=4,

The circle is drawn encircling the reference point (white) and the query points (black). The k-value is verified by adjusting in such a way as to optimize the indices.

Advantages and the Limitations

The model is capable of handling scenarios when the decision boundaries between classes are asymmetrical or irregular. Further, it does reduce the impact of a single noisy point, as it considers the votes of numerous neighbours. In addition, the model effectively handles circumstances involving incremental learning. It’s crucial to remember that the kNN model contains some drawbacks as it needs to figure out the distances between instances, and computationally expensiveness, especially with larger datasets. Furthermore, the decision made for k (number of neighbours) and the

distance metric can have an impact on how well kNN performs, hence it's important to choose a suitable value of k.

5.1.3 Gradient Boosting Classifier(GBC)

A powerful algorithm marked with significant importance in various applications, especially in terms of model accuracy over to the ability to capture complex relationships and interactions between features. The model is identical to the random forest, however, the GBC uses the principle of boosting with sequential training whilst the random forest uses the bagging principle with parallel training of the entire data set in place, where combining groups of classes is common. Yet it acts as an ensemble learning approach that combines multiple weak models of decision trees that iterates through by correcting the indicated errors caused by the previous model whilst reducing bias and variance. The process leads to improved generalization and prediction performances and elevates the model accuracy. Furthermore, the algorithm is known for handling heterogeneous data; numerical, categorical and text data with an inbuilt mechanism to handle the missing values with a mix of feature types showing versatility in applications. The model provides a measure of feature importance which is greatly important in feature selection and feature engineering and enhances the ability of interpretations. The iterative nature of the boosting process assists in mitigating the impact of noisy data and outliers. The imbalanced data is an issue in many cases. The GBC can resolve the significantly different classes by adding higher weight to the minority class during the training process. The hyperparameters like learning rate, depth of the tree, lambda value and a number of boosting iterations would allow the demanded customization and the fine-tuning of the model, despite the time taken (Baboota and Harleen Kaur 2019). The grid search was used for determining the hyperparameters (figure5.5).

5.1.4 Multi-layer Perceptron Classifier

The artificial Neural network (ANN) is a newer dimension and the research has been rooted back to the 1940s. A mathematical model was first introduced for the prediction of the football match outcome in the 1990s (Guan and X. Wang 2022). The Multi-layer Perceptron (MLP) Classifier is an artificial neural network that has been frequently used in specific work of predicting football match results effectively, as an ML task. MLP is another means of supervised learning impregnated with deep learning architecture where the input data is absorbed by the input layer, converted through some

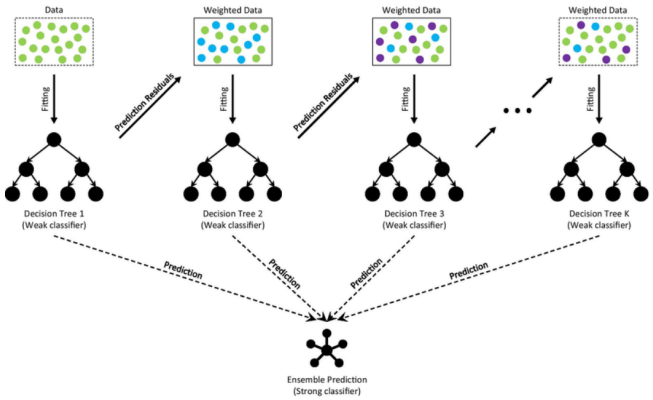


Figure 5.5: GBC Architecture, **https: //www.researchgate.net/figure/ The-architecture-of-Gradien**

hidden layers using applicable calculation functions and the result is evident through one output layer (Pugsee and Pattawong 2019). The algorithm is capable of learning complex patterns in the data, unlike any traditional models with non-linear relationships. Factors such as team strength, tactics, individual performances, and weather situations of a non-linear nature that are involved in real matches could be captured by the algorithm to showcase better performance. Also, it is capable of training large datasets by scaling up and identifying meaningful patterns and generating feature representations of higher dimensions. The model shows adequate flexibility in handling different data types including categorical features and reflects the capability of generalization through the learning patterns to make the predictions based on the trends and patterns.

Working Principle of MLP

Guan(2022) and Pugsee(2019) explained the training process of the MLP. Accordingly, backward propagation is the fundamental concept in the model's training process that allows the neural network to adjust their weights to minimize the difference between predicted outputs and actual target values. The input data is passed through the neural network layer by layer in the Forward Pass. Each neuron in the layer receives the input from the predecessor which computes the output with the aid of the weighted sum of inputs and the activation function to reveal the prediction through the output of the final layer. Compute Loss differentiates the predicted output from the actual target, which is reflected as the loss or the error. The Backward Pass(Backpropagation) determines the extent of contribution of each weight discussed above towards the error by working back reverse from the output to the input layer with the aid of establishing the gradient

of the respective loss, where the backpropagation leverages the chain rule of calculus to calculate the gradient of the respective loss. The activation function introduces the non-linearity to the network, realizing the complex relationships in data. Among the activation functions ”tanh” was used, as found effective through the grid search process. The learning rate determines the step size in the weight update process which involves subtracting the gradient of the loss with respect to a weight multiplied by the learning rate. However, the backpropagation may lead to overfitting as memorising the training data whilst looping back demands regularization or dropout layers to avoid. Gradient Descent uses the calculated gradient to update the weights of the network to mitigate the error and it is used as the optimizer by adjusting the weights based on their magnitude and the learning rate hyperparameters. Repeating the above steps with updating weights in iterative means for a number of epochs, realizes the convergence with better results. The MLP structure could be marked as below (figure5.6), where the input layer takes the input and the output layer gives away the output as process them through the hidden layer.

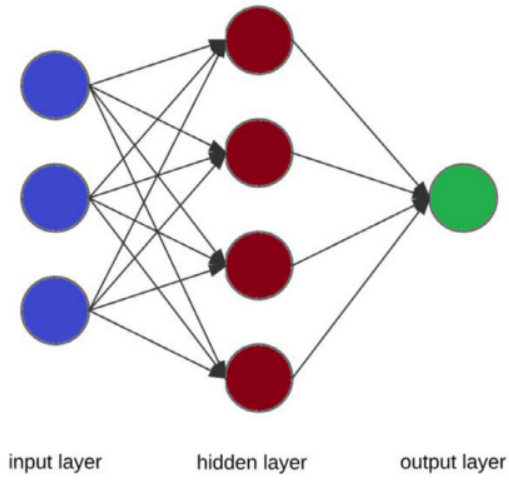


Figure 5.6: Multi-layer Perceptron Model,

5.1.5 Voting Classifier(VC)/Voter Ensemble (VE) Model

It is an ensemble learning method that combines multiple individual classifiers or models to make predictions (Maanijou et al. 2019). The model leverages the diversity to improve the overall predictive performances whilst enhancing the generalisation ability. It is embedded with a series of worthy attributes.

Robustness Maanijou(2019) states that it enhances robustness by reducing individual model prediction errors, through error correction with the aid of another model to ensure better accuracy.

Bias Reduction Its extended range of viewpoints does mitigate the bias of individual models' experiences during the training process and the assumptions made.

Enhanced Generalization The elevated generalisation characteristics shown towards the unseen data, by leveraging the collective knowledge of multiple models assisting to capture patterns of the data, compared to the individual models.

Mitigate Overfitting The combination of multiple models arrests the learning noises of individual models during training, resulting in stable and reliable predictions.

Model Diversity and Accuracy The model further works well with individual models of diverse strengths and weaknesses. Combining models with different algorithms, parameter settings, or training data subsets can help capture different aspects of the problem. The ensemble model does elevate accuracy compared to the individual model, (Gao et al. 2021).

Adaptability and Interpritability The model can be used with a variety of base models, making them versatile and applicable to different types of machine-learning tasks.The model transparency is high as simple and understandable.

5.2 Chapter Summary

The Methodology and the Model Design chapter reserves the flow adapted from the selection of the dataset to the model evaluation, including selecting the algorithms and training them as the core work followed by the preprocessing work dominated by feature engineering. Five algorithms were selected based on the empirical evidence provided. The RFC and GBC are tree-based algorithms where training on k-NN is based on a neighbour distance in making the predictions. The MLP is a different algorithm that works on deep neural network principles. The models were harnessed by VC/VE which follows the ensemble technique, in contrast.

Chapter 6

IMPLEMENTATION AND RESULTS

The models were trained primarily with 70 per cent of training data extracted from the last 10 years of EPL data (2013-2023) and validated with 20 per cent of it. However, the model accuracy observed primarily ranged just above 50 per cent where the highest accuracy was secured by GBC with 58 per cent, with significantly poor indices of draw predictions forcing to improve feature engineering works. Eventually, the data was re-sampled and finally secured with an elevated and marked performance of 66 per cent accuracy, followed by a strenuous effort of parameter tuning. The best model (VE) was tested against the unseen testing data, minimizing the perseverance for overfitting. The validation of the models was done through a few indices as explained below. The classification indices were mapped as A, D and H representing the Away Win, Draw, and the Home Win respectively.

Accuracy It is stated that the model accuracy is the ratio of correct predictions to the total number of predictions made by the model, (A. Kaur and I. Kaur 2018; Maanijou et al. 2019). It is expressed as a percentage, ranging from 0 per cent (completely inaccurate) to 100 per cent (completely accurate).

Precision Kaur(2018) and Maanijou(2019) stated that the precision gauges how accurately the model's optimistic predictions come true.

Recall/Sensitivity/TP Rate/ Probability of Detection Kaur(2018) and Maanijou(2019) referred to the sensitivity or true positive rate, which quantifies the percentage of real positive events that the model accurately detected.

```
Accuracy: 0.55
RFC Classification Report for validation data:
              precision    recall  f1-score   support

           A       0.55      0.60      0.57      1585
           D       0.38      0.12      0.18      1327
           H       0.58      0.79      0.67      2193

    accuracy                           0.55      5105
   macro avg       0.50      0.50      0.47      5105
weighted avg       0.52      0.55      0.51      5105
```

Figure 6.1: RFC Initial Base Model Performance,

F1-score Kaur(2018) and Maanijou(2019) state that the F1-score is the harmonic mean of recall and precision, which strikes a balance between the two metrics, When there is an imbalance in the class.

Confusion Matrix (CM) In the Confusion Matrix (CM), each row represents the actual class instances whereas each column represents instances of the predicted class.

6.1 Random Forest Model(RFC)

The base RFC base model was primarily trained and the results were moderate, despite the low precision, recall and the F1-score shown against draw predictions (figure6.1).

Explanation on RFC Base model How effectively the model predicts each class is shown by the accuracy, precision, recall, and F1-score (Maanijou et al. 2019). According to the categorization depicted, the model prediction differed substantially for classes D compared to A, and H. The model secured a validation accuracy of 55 per cent. The precision for class A is 0.55, indicating that 55 per cent of examples were correctly predicted. Class D and H have precision of 0.38 and 0.58 respectively. The Recall for class A is 0.60, which indicates that 60 per cent of all real instances of class A were correctly identified by the model. Recalls for classes D and H are 0.12 and 0.79, respectively. Class A, D, and H were able to secure F1 values of 0.57, 0.18, and 0.67, respectively. Accordingly, the model seems performed reasonably well for class H (Home team wins) with a high precision, recall and F1 score, with moderate performance in class A, despite the poor figures in draw predictions. In the Confusion Matrix (CM), the number of Away Win (A) prediction instances is 966/1585, the Draw prediction (D) instances are 106/1327, and the Home Win prediction (H) is 1802/2193,

Actual	Predicted Away Win	Predicted Draw	Predicted Home Win
Away Win	966	71	548
Draw	411	106	810
Home	339	52	1802

Table 6.1: Confusion Matrix for RFC Base model.

correctly made as shown in the table. This concludes the base model showing better Home win prediction, moderate Away win and worst Draw predictions, comparatively, demanding corrective actions Table 6.1).

Treating Class Imbalance If the dataset is imbalanced (that is, some classes have noticeably fewer samples than others), it is necessary to treat class imbalance by using methods like class weighting, oversampling, or under-sampling (Heijboer, Ong, and Zamberlan 2022). It is revealed that the lesser precision in prediction is due to the sample imbalance and tried off to apply under-sampling technique, in improving the Draw prediction. The below illustration is made on the under-sampling technique and yet the expected outcome at least over 50 has not been able to secured (figure6.2).

```
Under RS RFC Validation set report:
              precision    recall  f1-score   support

           A       0.54      0.62      0.58      1585
           D       0.36      0.34      0.35      1327
           H       0.65      0.60      0.62      2193

    accuracy                           0.54      5105
   macro avg       0.52      0.52      0.52      5105
weighted avg       0.54      0.54      0.54      5105

Under RS RFC Test set report:
              precision    recall  f1-score   support

           A       0.51      0.59      0.55       781
           D       0.33      0.30      0.31       653
           H       0.62      0.59      0.60      1081

    accuracy                           0.51      2515
   macro avg       0.49      0.49      0.49      2515
weighted avg       0.51      0.51      0.51      2515
```

Figure 6.2: UndersamplingRFC,

The Oversampling technique was used as a trial and error work to see whether any improvement was gained. The figures revealed that the oversampling even would not be able to harness the issue, considerably and would be tested for the combined sampling effectiveness ((figure6.3).

Over RS RFC Validation set report:

	precision	recall	f1-score	support
A	0.54	0.62	0.58	1585
D	0.37	0.28	0.32	1327
H	0.63	0.66	0.64	2193
accuracy			0.55	5105
macro avg	0.51	0.52	0.51	5105
weighted avg	0.53	0.55	0.54	5105

Over RS RFC Test set report:

	precision	recall	f1-score	support
A	0.52	0.59	0.55	781
D	0.32	0.24	0.27	653
H	0.61	0.65	0.63	1081
accuracy			0.52	2515
macro avg	0.48	0.49	0.49	2515
weighted avg	0.51	0.52	0.51	2515

Figure 6.3: OversamplingRFC,

Accuracy: 0.82
RFC Base Classification Report for validation data:

	precision	recall	f1-score	support
A	0.86	0.78	0.82	845
D	0.79	0.88	0.83	1022
H	0.89	0.73	0.80	297
accuracy			0.82	2164
macro avg	0.84	0.80	0.82	2164
weighted avg	0.83	0.82	0.82	2164

Figure 6.4: RFC SMOTE Model,

Finally, the Synthetic Minority Over Sampling Technique (SMOTE) indicated robust (Ghorbani and Ghousi 2020) and worked and this shows great improvement in all aspects in terms of predictions with 0.65 accuracy and other nearly equivalent indices.

6.1.1 Improved RFC Base Model After SMOTE Over Sampling

The hyperparameters were tuned as per the grid search showing some improvement in Draw precision (figure6.4).

The next step was hyperparameter tuning, in order to further elevate the prediction indices. The hyperparameters were tuned iterative manner with the base of the grid search results with respect to optimization of the RFC model and the process of work undertaken (figureB.10).

6.1.2 Best RFC Model

The hyperparameters were tuned as per the grid in optimizing the prediction with 0.66 accuracy and nearly equivalent indices in precision, recall, F1-score, micro, and weighted average for A, D, and H predictions (figure6.5). The Confusion Matrix (CM) does confirm the results obtained with the figures and the table shown (figure6.6). The same could be tabulated for ease of assimilation Table 6.2)

```
Accuracy:  0.654863
```

```
1 # Generate a classification report for validation data
2 print(" RFC Classification Report for Validation data:")
3 print(classification_report(y_val, rfc_best_pred2))
```

```
RFC Classification Report for Validation data:
              precision    recall  f1-score   support

           A       0.65      0.71      0.68      2193
           D       0.68      0.57      0.62      2193
           H       0.64      0.69      0.66      2194

    accuracy                           0.65      6580
   macro avg       0.66      0.65      0.65      6580
weighted avg       0.66      0.65      0.65      6580
```

Figure 6.5: RFC Best Model,

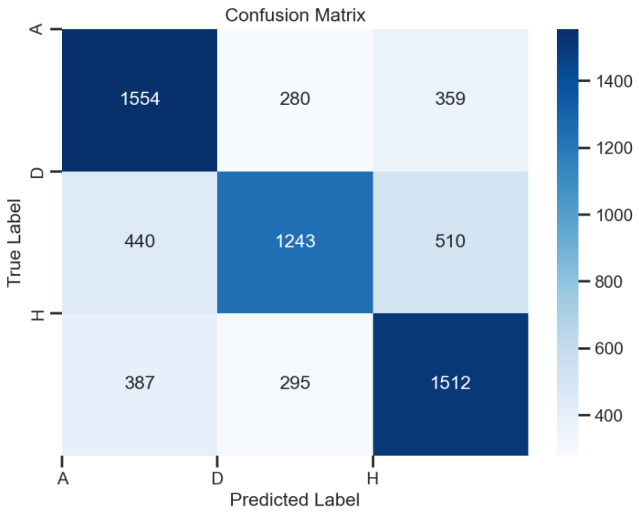


Figure 6.6: RFC CM,

Actual	Predicted Away Win	Predicted Draw	Predicted Home Win
Away Win	1554	280	359
Draw	440	1243	510
Home	387	295	1512

Table 6.2: Confusion Matrix for RFC model.

6.1.3 RFC Best Model Summary

The model secured a testing accuracy of 66 per cent higher figure with precision of 0.65, 0.68,0.64 whilst the Recall is 0.71, 0.57, 0.69, for classes A, D, and H respectively. The respective F1 scores are 0.68, 0.62, and 0.66. Accordingly, the model performs with high indices for all 3 classes(Home team wins, away team wins and draws) in terms of high accuracy, precision, recall and F1 score. In the Confusion Matrix (CM) marked, the number of correct Away Win (A) prediction instances of 1554/2193, the Draw prediction (D) instances of 1243/2193, and the Home Win prediction (H) instances of 1512/2193, as shown in the table. In addition, it shows over 66 per cent micro average and the weighted micro average figures against each index. This concludes the best RFC model outperforming with all-around better predictability, with respect to EPL football match results predictions.

6.2 k-Nearest Neighbors(kNN)

Primarily, the base kNN model was executed and got 53 per cent accuracy, demanding greater improvement, especially in draw predictions (figure6.7).

```
Accuracy: 0.55
Classification Report for validation data:
              precision    recall  f1-score   support

           A       0.53      0.54      0.53      1585
           D       0.33      0.18      0.23      1327
           H       0.57      0.73      0.64      2193

    accuracy                           0.53      5105
   macro avg       0.48      0.48      0.47      5105
weighted avg       0.50      0.53      0.50      5105
```

Figure 6.7: Base model knn,

6.2.1 Handling Sampling Imbalance

The SMOTE sampling was validated, finally to reveal the best model in the indices all around, with 58 per cent accuracy marked (figure6.8). The figures are verified through the Confusion Matrix(CM) and this shows great improvement in all aspects in terms of predictions (figure6.9). The figures were tabulated for ease of assimilation Table 6.3).

```
knn Best model Classification Report:
              precision    recall  f1-score   support

           A       0.60      0.64      0.62      2193
           D       0.53      0.57      0.55      2193
           H       0.63      0.55      0.59      2194

    accuracy                           0.58      6580
   macro avg       0.59      0.58      0.58      6580
weighted avg       0.59      0.58      0.58      6580
```

Figure 6.8: knn best model,

6.2.2 Optimization and Hyperparameter tuning

The hyperparameters were tuned iterative manner with the base of the grid search results with respect to optimization of the kNN model and a sample is depicted below to indicate the process, despite the ineffectiveness (figureB.11).

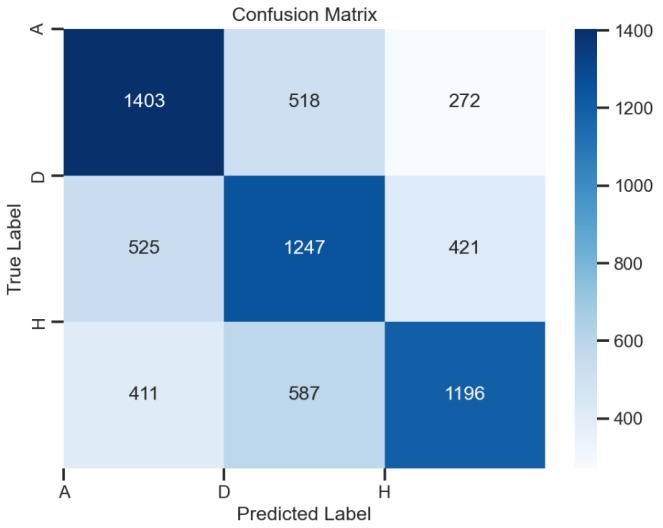


Figure 6.9: knn CM,

Actual	Predicted Away Win	Predicted Draw	Predicted Home Win
Away Win	1403	518	272
Draw	525	1247	421
Home	411	587	1196

Table 6.3: Confusion Matrix for kNN Best model

kNN Best Model Summary The model secured a validation accuracy of 58 per cent higher figure. The precision is 0.60, 0.53 and 0.63 whilst the Recall is 0.64, 0.57, and 0.55, for classes A, D, and H respectively. Class A, D, and H were able to secure F1 values of 0.62, 0.55, and 0.59, respectively. Accordingly, the model marks the highest predictability for class A (Away team wins) with a high accuracy, precision, recall and F1 score. As per the Confusion Matrix (CM), the number of correct Away Win (A) prediction instances is 1403/2193, the Draw prediction (D) instances are 1247/2193, and the Home Win prediction (H) is 1196/2193, made as shown in the table. In addition, it shows over 58 per cent micro average and the weighted micro average figures against each index. This concludes that the best k-NN model makes better all-around performance, in terms of football match results predictions, however, it rests below the performance of the RFC.

6.3 Gradient Boost Classifier (GBC)

The GBC model was trained without hyperparameters to reveal the difference and secure moderate performance compared to the base models with 58 per cent of accuracy, demanding treatment for class imbalance, over the lower prediction in draw results (figure6.10).

```
Accuracy: 0.58
GBC Base Classification Report:
              precision    recall  f1-score   support

           A       0.57      0.61      0.59      2193
           D       0.53      0.46      0.49      2193
           H       0.63      0.67      0.65      2194

    accuracy                           0.58      6580
   macro avg       0.58      0.58      0.58      6580
weighted avg       0.58      0.58      0.58      6580
```

Figure 6.10: GBC Base Model,

6.3.1 Optimization and Hyperparameter tuning through grid search

The hyperparameters were tuned through the grid search, despite the ineffectiveness shown (figureB.12).

6.3.2 Best GBC Model

The SMOTE re-sampling (CRS) technique was applied to improve the drawing figures and showed identical indices (figure6.11). The confusion matrix is used to verify the results (figure6.12). The figures were tabulated for ease of assimilation Table 6.4).

```
Accuracy:  0.581155
 Best GBC Classification Report for:
              precision    recall  f1-score   support

           A       0.57      0.61      0.59      2193
           D       0.53      0.46      0.49      2193
           H       0.63      0.67      0.65      2194

    accuracy                           0.58      6580
   macro avg       0.58      0.58      0.58      6580
weighted avg       0.58      0.58      0.58      6580
```

Figure 6.11: GBC (Best) Model,

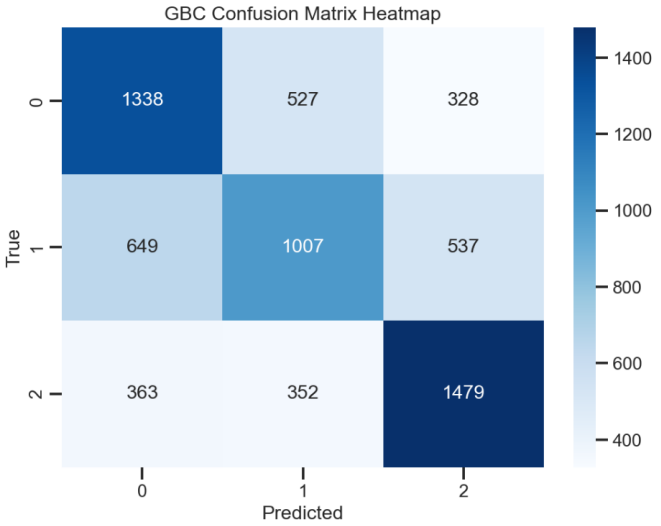


Figure 6.12: GBC-CM,

Actual	Predicted Away Win	Predicted Draw	Predicted Home Win
Away Win	1338	527	328
Draw	649	1007	537
Home	363	352	1479

Table 6.4: Confusion Matrix for GBC Best model

GBC Best Model Summary The model secured a moderate accuracy of 58 per cent with the Precision of 0.57, 0.53 and 0.63 whilst the Recall is 0.61, 0.46, and 0.67, for classes A, D, and H respectively. The Class A, D, and H were able to secure F1 values of 0.59, 0.49, and 0.65, respectively. As per the Confusion Matrix (CM), the number of correctly made Away Win (A) prediction instances is 1338/2193, the Draw prediction (D) instances are 1007/2193, and the Home Win prediction (H) is 1479/2193, as shown in the table. In addition, it shows 58 per cent micro average and the weighted micro average figures against each index. This concludes the best k-NN model performing moderately all-around performance, identical to kNN model.

6.4 Multi-Layer Perceptron(MLP)

The base MLP model was trained primarily and got the default results which are the lowest in base figures forcing to apply re-sampling.

6.4.1 Handling Sample Imbalance

Failing under-sampling, the SMOTE sampling was applied to find an alternative means of improvement and secured 61 per cent of accuracy (figure6.14). Hyperparameter tuning was done rigorously to improve the figures as an iterative process (figureB.13). Accordingly, the best model secured a validation accuracy of 63 per cent with Precision of 0.66, 0.59 and 0.64 and Recall of 0.66, 0.65, and 0.58 for classes A, D, and H, respectively. Class A, D, and H were able to secure F1 values of 0.66, 0.62, and 0.61, respectively. The CM was used to verify the model's accuracy.

```
Accuracy: 0.61
MLP Base Classification Report:
              precision    recall  f1-score   support

           A       0.62      0.65      0.64      2193
           D       0.59      0.61      0.60      2193
           H       0.62      0.57      0.59      2194

    accuracy                           0.61      6580
   macro avg       0.61      0.61      0.61      6580
weighted avg       0.61      0.61      0.61      6580
```

Figure 6.13: MLP (Base) Model,

6.4.2 Best MLP Model selection

The hyperparameter-tuned best model could be presented (figure6.14). The confusion matrix verifies the results (figure6.15). The SMOTE sample worked and showed great improvement in all aspects of predictions Table 6.5)..

Actual	Predicted Away Win	Predicted Draw	Predicted Home Win
Away Win	1450	428	315
Draw	378	1417	398
Home	379	553	1262

Table 6.5: Confusion Matrix for MLP Best model

Accuracy: 0.61
MLP Base Classification Report:

	precision	recall	f1-score	support
A	0.62	0.65	0.64	2193
D	0.59	0.61	0.60	2193
H	0.62	0.57	0.59	2194
accuracy			0.61	6580
macro avg	0.61	0.61	0.61	6580
weighted avg	0.61	0.61	0.61	6580

Figure 6.14: MLP (Base) Model,

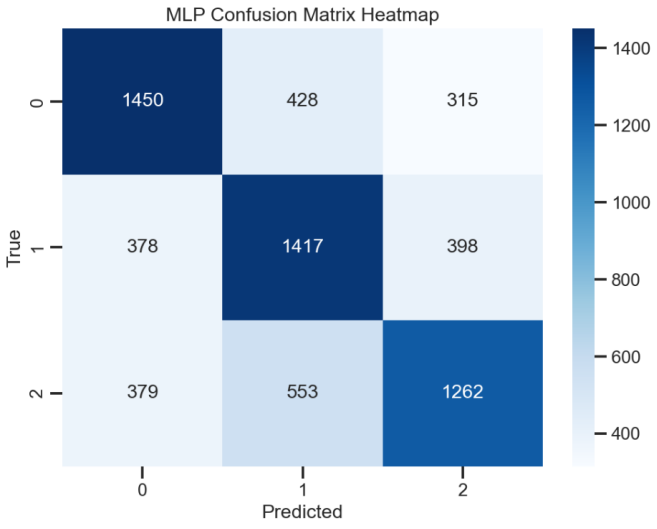


Figure 6.15: MLP-CM,

MLP Best Model Summary The model secured a validation accuracy of 63 per cent, just below the RFC figures with Precision of 0.66, 0.59 and 0.64 whilst the Recall 0.66, 0.65, and 0.58, for classes A, D, and H respectively. Class A, D, and H were able to secure F1 values of 0.66, 0.62, and 0.61, respectively. In the Confusion Matrix (CM), the number of Away Win (A) prediction instances is 1450/2193, the Draw prediction (D) instances are 1417/2193, and the Home Win prediction (H) is 1262/2193, correctly made as shown in the table. In addition, it shows 63 per cent micro average and the weighted micro average figures against each index. This concludes the best MLP model shows better all-around performance, in terms of football match results predictions that rest just below the RFC model.

6.5 Voter Ensemble

(Maanijou et al. 2019) emphasise that the Voter Ensemble model combines the multiple individual classifiers to improve overall predictive performance with better generalization performance and the figure reveals the outcomes of respective models. The figures show the model is outperforming (figure6.16). The figures could be verified through the confusion Matrix (figure6.17). The same could be tabulated for ease of assimilation The table, Table 6.6).

```
Accuracy:  0.649696
 Voting Classification Report for:
              precision    recall  f1-score   support

           A       0.67      0.69      0.68      2193
           D       0.62      0.64      0.63      2193
           H       0.67      0.62      0.64      2194

    accuracy                           0.65      6580
   macro avg       0.65      0.65      0.65      6580
weighted avg       0.65      0.65      0.65      6580
```

Figure 6.16: VC Model,

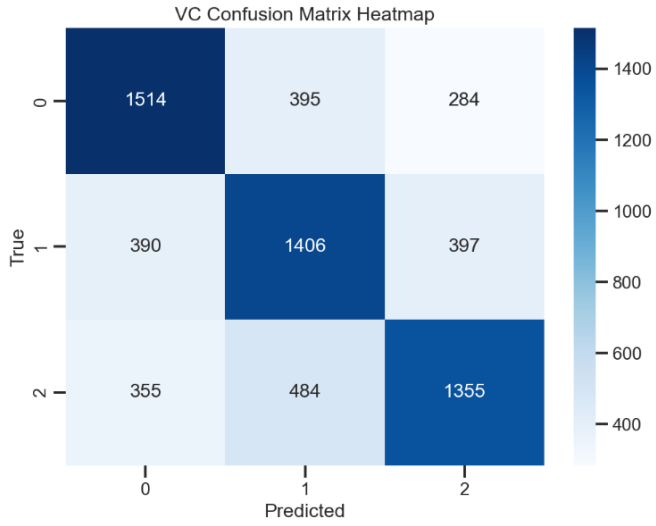


Figure 6.17: VC -CM,

Actual	Predicted Away Win	Predicted Draw	Predicted Home Win
Away Win	1514	395	284
Draw	390	1406	397
Home	355	484	1355

Table 6.6: Confusion Matrix for Best VC model

6.5.1 Voting Ensemble Best Model Summary

The model secured the marked validation accuracy of 65 per cent highest figure with precision of 0.67, 0.62, 0.67 whilst the Recall of 0.69, 0.64, and 0.62, for classes A, D, and H respectively. Class A, D, and H were able to secure F1 values of 0.68, 0.63, and 0.64, respectively. Accordingly, the model performs with the highest indices for away win predictions. In the CM, the number of correctly made Away Win (A) prediction instances is 1514/2193, the Draw prediction (D) instances are 1406/2193, and the Home Win prediction (H) is 1355/2193, as per the table. In addition, it shows 65 per cent micro average and the weighted micro average figures against each index. This concludes that the VE model makes the best all-around performance in terms of football match results predictions, over RFC.

6.6 Overall Best Model

Let's examine the best overall model! (figure**??**). It confirms that the RFC and VC mark the best identical model performance with 65 per cent accuracy.

Chapter Summary The 5 algorithms were used for the training process (where VC was used to harness the models) and the same were validated with validation data. Finally, the root indices headed by accuracy finalize the course. Accordingly, the RFC and the VC mark the highest identical accuracy of 66 per cent for the prediction of the football results. Hence, the best model will be selected during the evaluation process, after verification through the precision, Recall, F1-score, micro average along with other evaluation metrics.

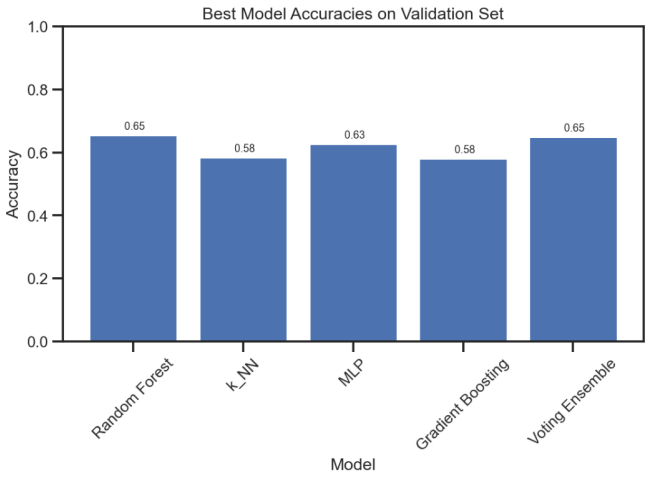


Figure 6.18: Comparision of Model Accuracy,

Chapter 7

MODEL EVALUATION

The evaluation process was rooted in splitting of the data into three subsets of data; training, validation and testing, where the training data is used for the model learning process and the resultant is validated with the validation data set. The final best models were once tested with the testing data set, to mitigate biases and overfitting.

7.1 Evaluation Matrix

It is stated in "Empirical Evaluation of Classification Algorithms for Fault Prediction in Open Source Projects", that Accuracy, Precision, Recall, and F1-score are a set of evaluation methods that could be used with the classification algorithms (**Maanijou et al.2019**; A. Kaur and I. Kaur 2018). In addition, a few more indices were used in the process; Cross-Validation (CV) Score, True vs. Predicted analysis, Receiver Operating Characteristic and Area Under the Curve(ROC/AUC) curves, and finally the SHAP analysis.

7.1.1 Accuracy

Kaur(2018) and Maanijou et al(2019) stated that the accuracy is termed as the overall level of correct prediction and could be represented below, Equations, 7.1, 7.2, 7.3, 7.4 are derived from **Wu et al.(2019)**. The TP is True Positive (number of correct predictions as positive) and TN (number of correct predictions as negative)is True Negative where FP (incorrectly predicted as positive)is False Positive and FN (incorrectly predicted as negative) is False Negative, respective to the predictions made.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{7.1}$$

7.1.2 Precision

Kaur(2018) and Maanijou(2019) states that the proportion of the TP in all positives,

$$\text{Precision} = \frac{TP}{TP + FP} \tag{7.2}$$

7.1.3 Recall/Sensitivity/TP Rate/ Probability of Detection(PD)

Kaur(2018) states that the Recall is the proportion of positive samples correctly identified. It is often referred to as sensitivity or true positive rate, quantifies the percentage of real positive events that the model reveals. The models that produce low PD are poor fault predictors.

$$[htbp]\text{Recall} = \frac{TP}{TP + FN} \tag{7.3}$$

7.1.4 F1-score

Kaur(2018) and Maanijou(2019) stated that the F1 score is the combination of the measurement of precision and the Recall.

$$[htbp]1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \tag{7.4}$$

7.1.5 Specificity

Kaur(2018) and Maanijou(2019), states that the specificity is the proportion of correctly identified negative cases (fault-free modules). The probability of a false alarm (PF) is a proportion of negative cases that are classified erroneously. The formula for $PF)$ is given by Equation 7.5. The lower values of Specificity increase the validation of fault-free modules since they would be incorrectly marked.

$$[htbp]PF = 1 - \text{Specificity} \tag{7.5}$$

7.1.6 Cross-Validation Score(CV)

Among the Monte Carlo re-sampling, Bootstrap, k-fold CV and repeated k-fold CV, the k-fold CV has been identified as a robust sampling technique (Moreno-Torres, Saez, and Herrera 2012), despite drawbacks in time-consuming. It mitigates the bias and avoids overfitting. Also named k-fold CV with (a mutually exclusive k-randomly split subset of data) equivalent in size where each model is trained and tested for k number of times. The (k-1) folds are used for the learning at each time whilst the last fold is used for the testing and the average of accuracy is taken by averaging individual accuracy components, despite the computational expensiveness inherent. Accordingly, the overall accuracy is represented by the summation of the average accuracy of each fold(A), Equation 7.6 is derived from Eryarsoy and Delen 2019.

$$[htbp]\text{CV} = \frac{1}{k}\sum\frac{A}{x} \tag{7.6}$$

Identifying The Best Model Let's identify the best model by validating through the testing data. Accordingly, RFC is lead with 66 per cent of accuracy on testing data (figure7.1).

7.2 Summary of Model Performances

Accuracy The summary of the performance of the models, in terms of accuracy, is depicted below and the RFC model outperformed with 0.66 per cent accuracy. Table 7.1).

Model	RFC	kNN	GBC	MLP	VE	Input
Acc.	0.66	0.58	0.58	0.63	0.65	1080

Table 7.1: Model Accuracy

7.2.1 Overall Performance with Cross Validation

The overall performance with 5-fold cross-validation could be represented here and how closely competing the RFC, MLP, VC and ultimately the RFC becomes the most robust model, considering overall indices (figure7.2).

```
f{name} ~ testing Accuracy: {accuracy_test: .2f}
Testing Classification Report:
              precision    recall  f1-score   support

           A       0.67      0.70      0.69      1080
           D       0.69      0.59      0.64      1081
           H       0.64      0.70      0.67      1080

    accuracy                           0.66      3241
   macro avg       0.67      0.66      0.66      3241
weighted avg       0.67      0.66      0.66      3241

f{name} ~ testing Accuracy: {accuracy_test: .2f}
Testing Classification Report:
              precision    recall  f1-score   support

           A       0.61      0.64      0.62      1080
           D       0.55      0.58      0.56      1081
           H       0.62      0.55      0.59      1080

    accuracy                           0.59      3241
   macro avg       0.59      0.59      0.59      3241
weighted avg       0.59      0.59      0.59      3241

f{name} ~ testing Accuracy: {accuracy_test: .2f}
Testing Classification Report:
              precision    recall  f1-score   support

           A       0.66      0.65      0.66      1080
           D       0.59      0.66      0.62      1081
           H       0.63      0.57      0.60      1080

    accuracy                           0.63      3241
   macro avg       0.63      0.63      0.63      3241
weighted avg       0.63      0.63      0.63      3241

f{name} ~ testing Accuracy: {accuracy_test: .2f}
Testing Classification Report:
              precision    recall  f1-score   support

           A       0.59      0.64      0.62      1080
           D       0.56      0.46      0.51      1081
           H       0.64      0.70      0.67      1080

    accuracy                           0.60      3241
   macro avg       0.60      0.60      0.60      3241
weighted avg       0.60      0.60      0.60      3241

f{name} ~ testing Accuracy: {accuracy_test: .2f}
Testing Classification Report:
              precision    recall  f1-score   support

           A       0.67      0.69      0.68      1080
           D       0.63      0.64      0.63      1081
           H       0.65      0.62      0.64      1080

    accuracy                           0.65      3241
   macro avg       0.65      0.65      0.65      3241
weighted avg       0.65      0.65      0.65      3241
```

Figure 7.1: Model Performance with CV,

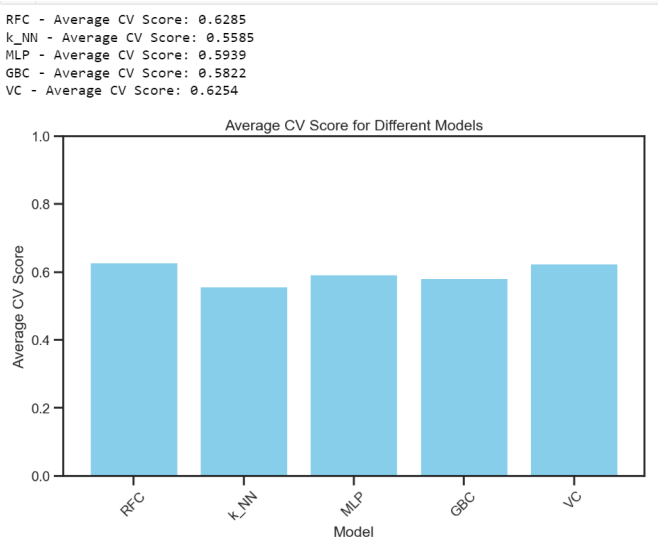


Figure 7.2: Average CV Score,

7.2.2 Other Indices of RFC Model

The table below showcases other outperforming figures of the RFC model Table **??**).

Predict.	Preci.	Recall	F1-score	Input
Away Win	0.67	0.70	0.69	1080
Draw	0.69	0.59	0.64	1081
Home	0.64	0.70	0.67	1080
Macro-Avg	0.67	0.66	0.88	1080
Weighted-Avg	0.67	0.66	0.66	1080

Table 7.2: Other Indices of Best model(RFC)

7.2.3 Receiver Operating Characteristic and Area Under the Curve(ROC-AUC) Analysis

Additionally, the ROC-AUC curves were drawn. It is stated that the ROC is formed by pairs of the Probability of False Alarm (PF) and the probability of Detection(PD) or Recall and is termed as a more general way for the performance measurement of the classifier, rather than configuring with numerical indices (A. Kaur and I. Kaur 2018). Kaur(2018) states that the AUC is commonly used for performance evaluation. In AUC, only the area with low PF and high PD is likely to have some impact on the performance of classifiers otherwise it is poor performance. The great advantage posed by the curve is its robustness towards imbalanced class distributions. Accordingly, the figure depicts the mean ROC curve (AUC of 0.95 per cent), the trade-off between the True Positive Rate(TPR) and the False Positive Rate (FPR) with respect to the away win prediction as 94 per cent, the trade-off between the True Positive Rate(TPR) and the False Positive Rate (FPR) with respect to the Draw prediction of 94 per cent, and the trade-off between the True Positive Rate(TPR) and the False Positive Rate (FPR) with respect to the Home Win prediction of 98 per cent(ROC class mapping = 'A': 0, 'D': 1, 'H': 2). Further, the area under the curve (AUC) summarizes the model's overall performance. The higher AUC indicates better discrimination between the 3 classes. Let's Verify the best VC model with ROC-AUC Scores,(figure7.3).

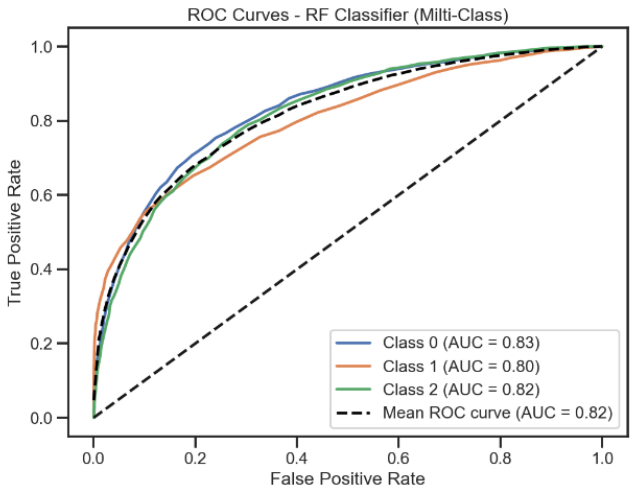


Figure 7.3: ROC-AUC Curve of Best RFC Modell,

7.2.4 True vs Predicted Analysis with the best model, RFC

The True vs. Predicted analysis was done to compare the actual vs. prediction that reveals the model performance and accuracy, which gives insights into how well the model can discriminate between the classes and help in making decisions about the models' threshold values for class assignments. Let's verify the best Voting Classifier Ensemble model with True vs. Predicted analysis that marks over 85 per cent overall accuracy, precision, Recall and the F1-score., showing the model's robustness to be used for future applications (figure7.4).

```
Confusion Matrix:
<function confusion_matrix at 0x000002C6BAEB7550>
Accuracy: 0.6548632218844985
Precision: 0.6548632218844985
Recall: 0.6548632218844985
F1 Score: 0.6548632218844985
```

Figure 7.4: True Vs Predicted Analysis of RFC Best Model,

7.2.5 SHapley Additive exPlanations (SHAP) Analysis

The SHAP summary and force plots are powerful visualization tools commonly used for interpreting feature importance (Moustakidis et al. 2023; Garnica-Caparrós and Memmert 2021). In this context, the feature contribution for the home win (H), Draw (D) and Away win (A) predictions could be plotted with SHAP summary plots to reveal their respective importance. In addition, the SHAP values explain how each feature value affects the model's prediction for the specific instance as how team statistics affect the model outcome, enabling better model understanding. Further, the SHAP values provide insights into the direction and magnitude of the feature's effect on predictions.

Away Win Feature Importance The feature importance in away wins (A) shows the importance of a number of away shots on target, taking corners, away shots, away win betting odds, and Home Lost Streak influencing the goal accomplishment (figure7.5).

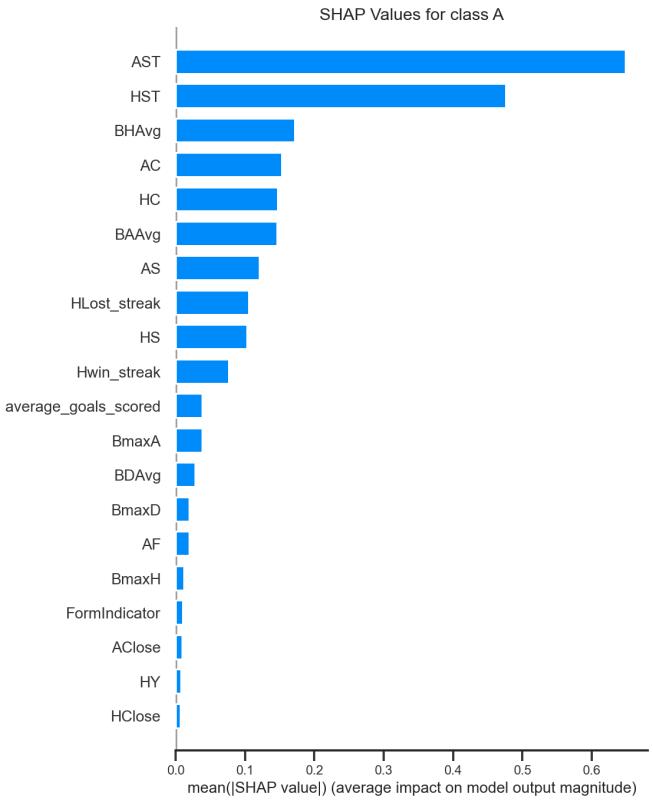


Figure 7.5: SHAP Symmary Plots for Away Win,

Home Win Feature Importance The feature importance in home wins (H) shows the importance of the number of home shots on target, taking corners, home shots, average home win betting odds, and average goals scored towards the goal accomplishment (figure7.6).

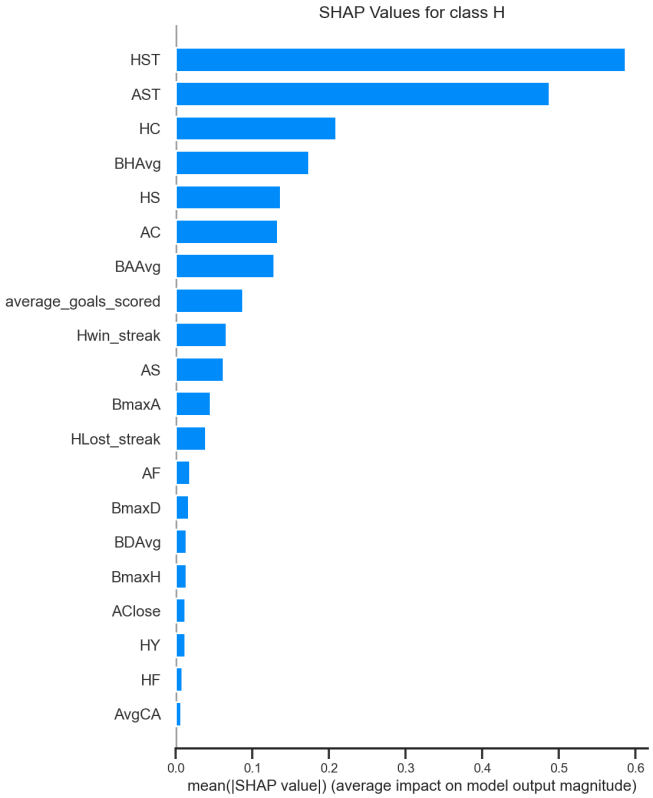


Figure 7.6: SHAP Symmary Plots for Home Win,

Feature Importance for Drawing the Match The feature importance in draws (D) shows the importance of Draw betting Odds, average goals scored and even average away win odds towards the goal accomplishment (figure7.7). All three classes could be depicted in the same plot for ease of assimilation (figure7.8).

As per the set objectives, let's verify compliance with the below objectives.

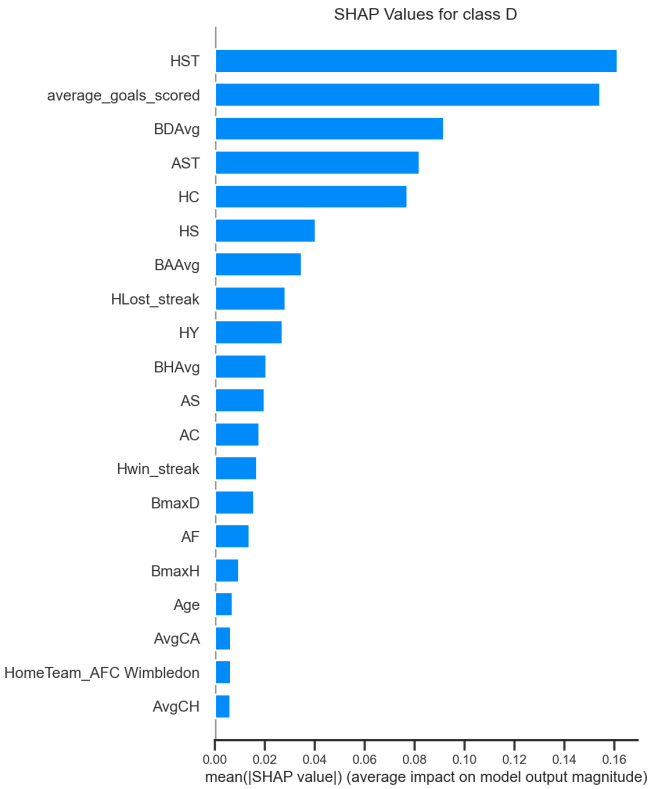


Figure 7.7: SHAP Symmary Plots for Draw,

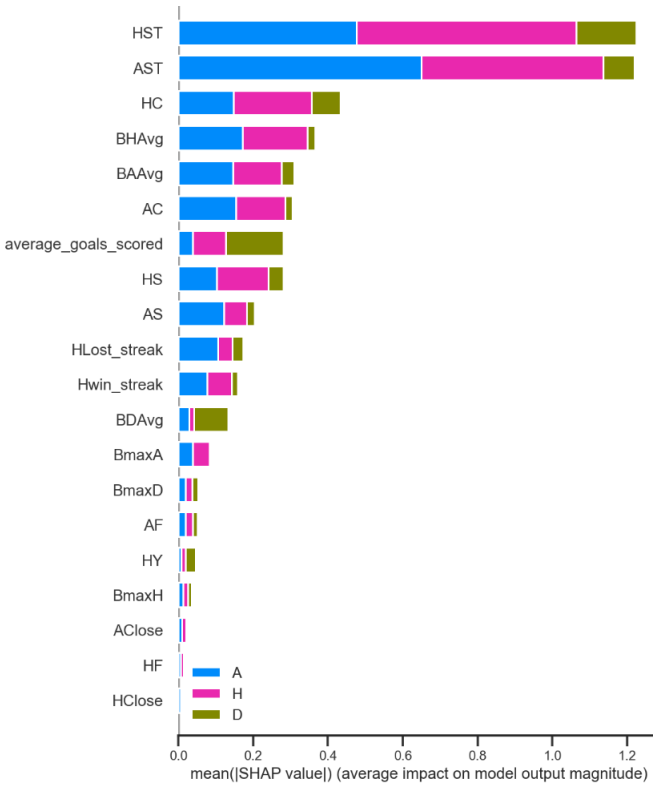


Figure 7.8: SHAP Cobined Summary Plot,

7.3 Assessing Whether Selected Appropriate dataset, Understood the tasks and Features

Alfredo(2019) stated EPL football has been the most played and watched game in the world. Further, the last 10 years of data amounting to just over 25398 observations were considered for the analysis and training input is 10912. Hence, together with the literature, it is concluded that the most appropriate data has been selected for the prediction process.

7.3.1 Asses the Workmanship in Processing the data

The workmanship followed the popular eventualities including the replacement of missing values, encoding categorical data, feature engineering, getting ELO rating, clustering, feature selection, and feature scaling prior to re-sampling to equalize the input prior to training data through scientifically selected algorithms that follow diverse mechanisms of tree-based, neighbour based, neural networks, ensemble algorithms with 70 per cent of data (training), as the quota majority followed. Further, all 5 models were performed with elevated indices above 58 per cent accuracy and equivalent other indices, conforming the process.

7.3.2 Assess whether selected potentially contributed features

All 5 models secure over 58 per cent model accuracy with the selected features by the SelectKBest model with the assistance of the intuition given by the correlation matrix and the threshold number of components enforced through PCA. The cross-validation scores of the best model stand over 66 per cent and this is a marked figure as per the literature stated. Hence, it could be concluded that the feature selections were done with great care, despite the fact that continual improvements could be evident, over the nature of the software upgrade and the competitiveness with investors.

7.3.3 Assess whether ML models were trained for the prediction of the football match results, using the selected features of the dataset.

The re-sampled data was fed for RFC, kNN, GBC, MLP, and VE models with 70 per cent of training data for training the models, using the selected features, as per the

Jupiter notebook submitted herewith. The models have secured over 59 per cent of model accuracy with sound other indices, indicating compliance.

7.3.4 Evaluate whether Validate the prediction with the actual football match results

Eventually, the trained models were validated using the 20 per cent of validation data and finally, the best model was revealed through hyperparameter tuning. They were tested using the 10 per cent testing data. The evaluation process included CM, CV, ROC-AUC curves and true vs. predicted analysis. The SHAP analysis was done for global explanations. Finally achieved a marked figure of 66 per cent accuracy (well supported by other indices; precision, Recall and F1 score, CV, ROC-AUC) for the best model, RFC. The rigorous process of verification confirms the model's robustness. Hence, it is verified in a few dimensions that the data processing has been carried out with credentials.

7.4 Evaluation Summary

The 5 models; RFC, kNN, GBC, MLP and VC/ VE were tested using the testing data and revealed that the RFC outperformed with 66 per cent accuracy with near other indices; precision (of 0.67, 0.69 and 0.64), Recall (of 0.70, 0.59, and 0.70) and F1-score(0.69, 0.64, and 0.67), for classes A, D, and H, respectively. The models were eventually cross-validated to arrest any kind of biasing if any and yet shown the RFC as the most outperforming model. Apart from the verification done through the CM, the model RFC showed over 66 per cent micro average and the weighted micro average figures against each index. In addition, the models were evaluated with the aid of ROC-AUC, True vs. Prediction for getting the model consistency in terms of robustness. AUC represents 0.82 would be a sound figure. The SHAP analysis was carried out to reveal the feature importance in the respective match outcome predictions and to make a clear interpretation, finally. Eventually, it concludes that the Random Forest Classifier model secures the best all-around performances, in terms of football match results predictions and could be recommended as a robust model for future applications.

Chapter 8

ITERATIVE IMPROVEMENTS

The iterative improvements are done throughout the process to improve the model prediction indices, that extended from diverse perspectives. The literature, Feature Engineering and Feature Selection processes were iteratively amended to secure better accuracy while sealing other indices intact, as handled the Sampling Imbalance with SMOTE. The introduction of the Voter Ensemble brought great competitiveness. Eventually, RFC secured 66 per cent model accuracy on testing data, finally.

Chapter 9

CONCLUSION AND FUTURE WORKS

Research on the prediction of football match results has been undertaken using the EPL data for 10 years from 2013. The endeavour was to establish a robust model for match results prediction, using an appropriate dataset. EPL data set was selected as the most played and watched game. The data extraction, familiarization, and missing value imputation were done as preprocessing work. In addition, feature engineering works were undertaken to get the most appropriate subset of data in view of improving the predictability of match results. The correlation matrix was used in identifying the independent features as needed for the model training prior to selecting the most contributory 21 feature. Scaling features and encoding Categorical features were done eventually followed by investigating the contribution of clustering and application of ELO rating, prior to splitting the data for the model application. The selected five models were trained using 70 per cent of training data, and results were validated with 20 per cent of validation data. Eventually, the best models were tested with testing data, minimising the data leakage. The process was evaluated to find out the match results through the prediction based on Accuracy, Precision, Recall, F1-score, CV scores, ROC-AUC curves, and True vs. Predicted Analysis. Accordingly, the RFC model showed a marked accuracy of 66 per cent with nearly equivalent other indices, preserving model application for the prediction of future match results. Eventually, the SHAP analysis was integrated, in replicating the feature importance towards each class of match outcome.

FUTURE WORK The computational complexity affected the process over the limited capacities of the personal computer to reveal the results within the allotted time frame. This includes restricting the EPL data for the last ten years, and the number of feature engineering processes as excluded time-consuming processes whilst selecting algorithms with moderate performance (avoiding sophisticated models such as advanced neural networks as time-consuming). However, this is an endless process, with the contemporary improvement in software development and excitement with the investors, especially. Accordingly, it is recommended below points for future proposals, as already referred to in the report,

- Select a larger, diverse dataset for the processing to reveal more robust model performances.
- Consider diverse features alike, player rankings, injuries to the player, substitute players, and weather conditions of the venues.
- Integrate match news to improve the recommendation

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Appendix A

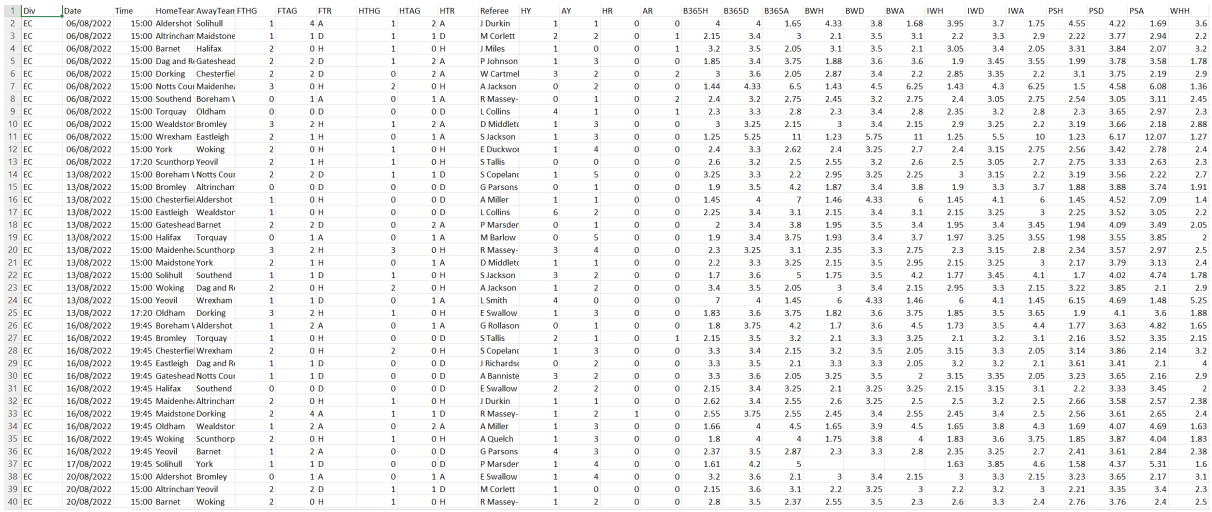
TOR

Department of Computing and Mathematics Computing and Digital Technology Postgraduate Programmes Terms of Reference Coversheet	
Student Name:	Chathura Chinthaka Morakanda Gamage
University I.D.:	22542695
Academic Supervisor:	Xia Cui
External Collaborator (optional):	Not Applicable (N/A)
Project Title:	Predicting the Results of Football Matches with machine learning models
EthOS Reference Number	57278
Degree Title:	MSc Data Science
Project Unit Code:	6G7V0007
Credit Rating:	60
Start date:	01.06.2023
ToR date:	27.06.2023
Intended Submission Date	21.09.2023
Signature and date student	 27.06.2023
Signature and date external	N/A

Figure A.1: TOR,

Appendix B

Cross-Section of the Dataset and Code Snipets



	Div	Date	Time	HomeTeam	AwayTeam	FTHG	FTAG	FTR	HTHG	HTAG	HTR	Referee	HY	AY	HR	AR	B365H	B365D	B365A	BWH	BWD	BWA	IWH	IWD	IWA	PSH	PSD	PSA	WHH
2	EC	06/08/2022	15:00	Aldershot	Solihull	1	4	A	1	2	A	J Durkin	1	1	0	0	4	4	1.65	4.33	3.8	1.68	3.95	3.7	1.75	4.55	4.22	1.69	3.6
3	EC	06/08/2022	15:00	Altrincham	Maidstone	1	1	D	1	1	D	M Corlett	2	2	0	1	2.15	3.4	3	2.1	3.5	3.1	2.2	3.3	2.9	2.22	3.77	2.94	2.2
4	EC	06/08/2022	15:00	Barnet	Halifax	2	0	H	1	0	H	J Miles	1	0	0	1	3.2	3.5	2.05	3.1	3.5	2.1	3.05	3.4	2.05	3.31	3.84	2.07	3.2
5	EC	06/08/2022	15:00	Dag and Rd	Gateshead	2	2	D	1	2	A	P Johnson	1	3	0	0	1.85	3.4	3.75	1.88	3.6	3.6	1.9	3.45	3.55	1.99	3.78	3.58	1.78
6	EC	06/08/2022	15:00	Dorking	Chesterfie	2	2	D	0	2	A	W Cartmel	3	2	0	2	3	3.6	2.05	2.87	3.4	2.2	2.85	3.35	2.2	3.1	3.75	2.19	2.9
7	EC	06/08/2022	15:00	Notts Cour	Maidenhe	3	0	H	2	0	H	A Jackson	0	2	0	0	1.44	4.33	6.5	1.43	4.5	6.25	1.43	4.3	6.25	1.5	4.58	6.08	1.36
8	EC	06/08/2022	15:00	Southend	Boreham \	0	1	A	0	1	A	R Massey-	0	1	0	2	2.4	3.2	2.75	2.45	3.2	2.75	2.4	3.05	2.75	2.54	3.05	3.11	2.45
9	EC	06/08/2022	15:00	Torquay	Oldham	0	0	D	0	0	D	L Collins	4	1	0	1	2.3	3.3	2.8	2.3	3.4	2.8	2.35	3.2	2.8	2.3	3.65	2.97	2.3
10	EC	06/08/2022	15:00	Wealdstor	Bromley	3	2	H	1	2	A	D Middlet	1	3	0	0	3	3.25	2.15	3	3.4	2.15	2.9	3.25	2.2	3.19	3.66	2.18	2.88
11	EC	06/08/2022	15:00	Wrexham	Eastleigh	2	1	H	0	1	A	S Jackson	1	3	0	0	1.25	5.25	11	1.23	5.75	11	1.25	5.5	10	1.23	6.17	12.07	1.27
12	EC	06/08/2022	15:00	York	Woking	2	0	H	1	0	H	E Duckwo	1	4	0	0	2.4	3.3	2.62	2.4	3.25	2.7	2.4	3.15	2.75	2.56	3.42	2.78	2.4
13	EC	06/08/2022	17:20	Scunthorp	Yeovil	2	1	H	1	0	H	S Tallis	0	0	0	0	2.6	3.2	2.5	2.55	3.2	2.6	2.5	3.05	2.7	2.75	3.33	2.63	2.3
14	EC	13/08/2022	15:00	Boreham \	Notts Cour	2	2	D	1	1	D	S Copelanc	1	5	0	0	3.25	3.3	2.2	2.95	3.25	2.25	3	3.15	2.2	3.19	3.56	2.22	2.7
15	EC	13/08/2022	15:00	Bromley	Altrincham	0	0	D	0	0	D	G Parsons	0	1	0	0	1.9	3.5	4.2	1.87	3.4	3.8	1.9	3.3	3.7	1.88	3.88	3.74	1.91
16	EC	13/08/2022	15:00	Chesterfie	Aldershot	1	0	H	0	0	D	A Miller	1	1	0	0	1.45	4	7	1.46	4.33	6	1.45	4.1	6	1.45	4.52	7.09	1.4
17	EC	13/08/2022	15:00	Eastleigh	Wealdstor	1	0	H	0	0	D	L Collins	6	2	0	0	2.25	3.4	3.1	2.15	3.4	3.1	2.15	3.25	3	2.25	3.52	3.05	2.2
18	EC	13/08/2022	15:00	Gateshead	Barnet	2	2	D	0	2	A	P Marsder	0	1	0	0	2	3.4	3.8	1.95	3.5	3.4	1.95	3.4	3.45	1.94	4.09	3.49	2.05
19	EC	13/08/2022	15:00	Halifax	Torquay	0	1	A	0	1	A	M Barlow	0	5	0	0	1.9	3.4	3.75	1.93	3.4	3.7	1.97	3.25	3.55	1.98	3.55	3.85	2
20	EC	13/08/2022	15:00	Maidenhe	Scunthorp	3	2	H	3	0	H	R Massey-	3	4	0	0	2.3	3.25	3.1	2.35	3.3	2.75	2.3	3.15	2.8	2.34	3.57	2.97	2.5
21	EC	13/08/2022	15:00	Maidstone	York	2	1	H	0	1	A	D Middlet	1	1	0	0	2.2	3.3	3.25	2.15	3.5	2.95	2.15	3.25	3	2.17	3.79	3.13	2.4
22	EC	13/08/2022	15:00	Solihull	Southend	1	1	D	1	0	H	S Jackson	3	2	0	0	1.7	3.6	5	1.75	3.5	4.2	1.77	3.45	4.1	1.7	4.02	4.74	1.78
23	EC	13/08/2022	15:00	Woking	Dag and Rd	2	0	H	2	0	H	A Jackson	1	2	0	0	3.4	3.5	2.05	3	3.4	2.15	2.95	3.3	2.15	3.22	3.85	2.1	2.9
24	EC	13/08/2022	15:00	Yeovil	Wrexham	1	1	D	0	1	A	L Smith	4	0	0	0	7	4	1.45	6	4.33	1.46	6	4.1	1.45	6.15	4.69	1.48	5.25
25	EC	13/08/2022	17:20	Oldham	Dorking	3	2	H	1	0	H	E Swallow	1	3	0	0	1.83	3.6	3.75	1.82	3.6	3.75	1.85	3.5	3.65	1.9	4.1	3.6	1.88
26	EC	16/08/2022	19:45	Boreham \	Aldershot	1	2	A	0	1	A	G Rollason	0	1	0	0	1.8	3.75	4.2	1.7	3.6	4.5	1.73	3.5	4.4	1.77	3.63	4.82	1.65
27	EC	16/08/2022	19:45	Bromley	Torquay	1	0	H	0	0	D	S Tallis	2	1	0	1	2.15	3.5	3.2	2.1	3.3	3.25	2.1	3.2	3.1	2.16	3.52	3.35	2.15
28	EC	16/08/2022	19:45	Chesterfie	Wrexham	2	0	H	2	0	H	S Copelanc	1	3	0	0	3.3	3.4	2.15	3.2	3.5	2.05	3.15	3.3	2.05	3.14	3.86	2.14	3.2
29	EC	16/08/2022	19:45	Eastleigh	Dag and Rd	1	1	D	0	0	D	J Richards	0	2	0	0	3.3	3.5	2.1	3.3	3.3	2.05	3.2	3.2	2.1	3.61	3.41	2.1	4
30	EC	16/08/2022	19:45	Gateshead	Notts Cour	1	1	D	0	0	D	A Banniste	3	2	0	0	3.3	3.6	2.05	3.25	3.5	2	3.15	3.35	2.05	3.23	3.65	2.16	2.9
31	EC	16/08/2022	19:45	Halifax	Southend	0	0	D	0	0	D	E Swallow	2	2	0	0	2.15	3.4	3.25	2.1	3.25	3.25	2.15	3.15	3.1	2.2	3.33	3.45	2
32	EC	16/08/2022	19:45	Maidenhe	Altrincham	2	0	H	1	0	H	J Durkin	1	1	0	0	2.62	3.4	2.55	2.6	3.25	2.5	2.5	3.2	2.5	2.66	3.58	2.57	2.38
33	EC	16/08/2022	19:45	Maidstone	Dorking	2	4	A	1	1	D	R Massey-	1	2	1	0	2.55	3.75	2.55	2.45	3.4	2.55	2.45	3.4	2.5	2.56	3.61	2.65	2.4
34	EC	16/08/2022	19:45	Oldham	Wealdstor	1	2	A	0	2	A	A Miller	1	3	0	0	1.66	4	4.5	1.65	3.9	4.5	1.65	3.8	4.3	1.69	4.07	4.69	1.63
35	EC	16/08/2022	19:45	Woking	Scunthorp	2	0	H	1	0	H	A Quelch	1	3	0	0	1.8	4	4	1.75	3.8	4	1.83	3.6	3.75	1.85	3.87	4.04	1.83
36	EC	16/08/2022	19:45	Yeovil	Barnet	1	2	A	0	0	D	G Parsons	4	3	0	0	2.37	3.5	2.87	2.3	3.3	2.8	2.35	3.25	2.7	2.41	3.61	2.84	2.38
37	EC	17/08/2022	19:45	Solihull	York	1	1	D	0	0	D	P Marsder	1	4	0	0	1.61	4.2	5				1.63	3.85	4.6	1.58	4.37	5.31	1.6
38	EC	20/08/2022	15:00	Aldershot	Bromley	0	1	A	0	1	A	E Swallow	1	4	0	0	3.2	3.6	2.1	3	3.4	2.15	3	3.3	2.15	3.23	3.65	2.17	3.1
39	EC	20/08/2022	15:00	Altrincham	Yeovil	2	2	D	1	1	D	M Corlett	1	0	0	0	2.15	3.6	3.1	2.2	3.25	3	2.2	3.2	3	2.21	3.35	3.4	2.3
40	EC	20/08/2022	15:00	Barnet	Woking	2	0	H	1	0	H	R Massey-	1	2	0	0	2.8	3.5	2.37	2.55	3.5	2.3	2.6	3.3	2.4	2.76	3.76	2.4	2.5

Figure B.1: Cross Section of the Data,



```
# Home win Odds(as BHAvg)
cols_to_average = ['B365H', 'BWH','IWH', 'SJH', 'LBH', 'WHH', 'VCH', 'PSH', 'PSCH', 'MaxH', 'AvgH', 'B365AHH', 'PAHH']

fb_df['BHAvg'] = fb_df[cols_to_average].mean(axis=1)
```

```
# Draw Odds (as BDAvg)
cols_to_average = ['B365D', 'BWD', 'IWD', 'SJD', 'LBD', 'WHD', 'VCD', 'PSD', 'PSCD', 'MaxD', 'AvgD', 'B365CD']
fb_df['BDAvg'] = fb_df[cols_to_average].mean(axis=1)

```

```
# Away Win Odds (as BAAvg)
cols_to_average = ['B365A','BWA', 'IWA', 'SJA', 'LBA', 'WHA', 'VCA','PSA', 'PSCA', 'MaxA', 'AvgA', 'B365AHA', 'PAHA']
fb_df['BAAvg'] = fb_df[cols_to_average].mean(axis=1)
```

```
# Averaging  Max Home Odds (as BmaxH)
cols_to_average = ['BbAvAHH', 'BbAvH','AvgAHH']
fb_df['BmaxH'] = fb_df[cols_to_average].mean(axis=1)
```

```
# Averaging  Max Home Odds (as BmaxH)
# cols_to_average = ['BbMxH',  'BbMxAHH', 'MaxAHH']
# fb_df['BmaxH'] = fb_df[cols_to_average].mean(axis=1)
```

```
# Averaging  Max Draw Odds (as BmaxD)
cols_to_average = ['AvgCD', 'BbAvD']
fb_df['BmaxD'] = fb_df[cols_to_average].mean(axis=1)
```

```
# # Averaging  Max Draw Odds (as BmaxD)
# cols_to_average = ['BbMxD', 'MaxCD']
# fb_df['BmaxD'] = fb_df[cols_to_average].mean(axis=1)
```

```
# Averaging  Max Away Odds (as BmaxA)
cols_to_average = ['BbAvAHA','BbAvA', 'AvgAHA']
fb_df['BmaxA'] = fb_df[cols_to_average].mean(axis=1)
```

```
# # Averaging  Max Away Odds (as BmaxA)
# cols_to_average = ['BbMxA', 'BbMxAHA','MaxAHA']
# fb_df['BmaxA'] = fb_df[cols_to_average].mean(axis=1)
```

```
# Max Betting Odds  Over 2.5 goals (as 'MaxB>2.5')
cols_to_average = ['BbMx>2.5', 'B365>2.5', 'P>2.5', 'Max>2.5', 'B365C>2.5','PC>2.5', 'MaxC>2.5',  'BbAv>2.5','Avg>2.5', 'Avg
fb_df['MaxB>2.5'] = fb_df[cols_to_average].mean(axis=1)
```

```
# Max Betting Odds Below 2.5 goals (as 'Bo<2.5')
cols_to_average = ['BbMx<2.5',  'B365<2.5', 'P<2.5', 'Max<2.5',  'B365C<2.5', 'PC<2.5', 'MaxC<2.5', 'BbAv<2.5', 'AvgC<2.5',
fb_df['Bo<2.5'] = fb_df[cols_to_average].mean(axis=1)
```

Figure B.2: Shrinking Betting Odds,



3.2..2. Derive new features using existing to support the prediction

```
# Gwet the gap of the goals acquired and conceeded
fb_df['goal_gap'] = fb_df['FTHG']-fb_df['FTAG']
```

```
C:\Users\chint\AppData\Local\Temp\ipykernel_20028\3901712335.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  fb_df['goal_gap'] = fb_df['FTHG']-fb_df['FTAG']
```

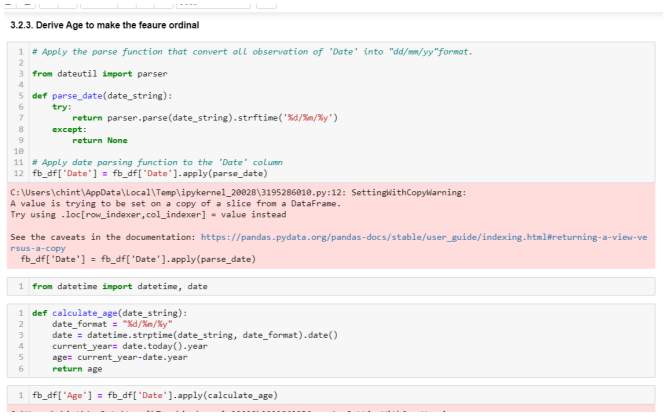
```
# Lets check on winning streak of the teams making respective cumilative sums
fb_df ['Hwin_streak'] = (fb_df['FTR'] == 'H').astype(int).groupby(fb_df['HomeTeam']).cumsum()
fb_df ['HLost_streak'] = (fb_df['FTR'] == 'A').astype(int).groupby(fb_df['AwayTeam']).cumsum()
```

```
C:\Users\chint\AppData\Local\Temp\ipykernel_20028\4039966409.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  fb_df ['Hwin_streak'] = (fb_df['FTR'] == 'H').astype(int).groupby(fb_df['HomeTeam']).cumsum()
C:\Users\chint\AppData\Local\Temp\ipykernel_20028\4039966409.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  fb_df ['HLost_streak'] = (fb_df['FTR'] == 'A').astype(int).groupby(fb_df['AwayTeam']).cumsum()
```

Figure B.3: Deriving New Features,



3.2.3. Derive Age to make the feaure ordinal

```
# Apply the parse function that convert all observation of 'Date' into "dd/mm/yy"format.

from dateutil import parser

def parse_date(date_string):
    try:
        return parser.parse(date_string).strftime('%d/%m/%y')
    except:
        return None

# Apply date parsing function to the 'Date' column
fb_df['Date'] = fb_df['Date'].apply(parse_date)
```

```
C:\Users\chint\AppData\Local\Temp\ipykernel_20028\3195286010.py:12: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

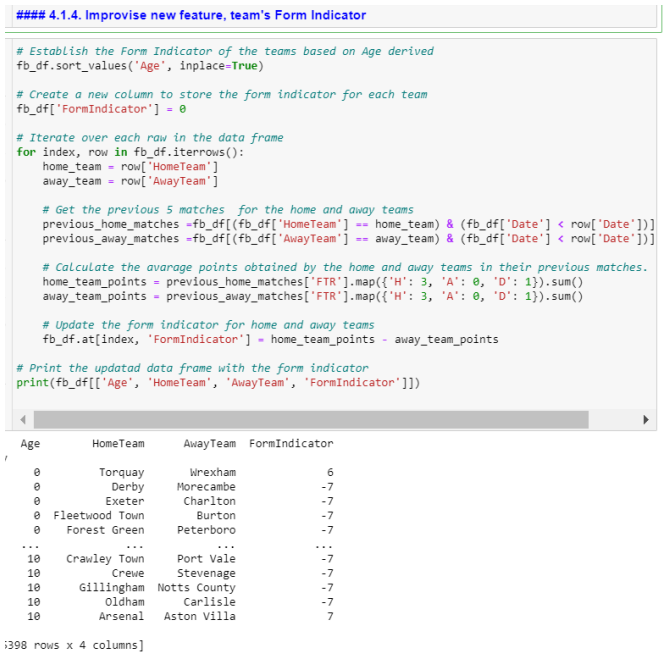
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  fb_df['Date'] = fb_df['Date'].apply(parse_date)
```

```
from datetime import datetime, date
```

```
def calculate_age(date_string):
    date_format = "%d/%m/%y"
    date = datetime.strptime(date_string, date_format).date()
    current_year= date.today().year
    age= current_year-date.year
    return age
```

```
fb_df['Age'] = fb_df['Date'].apply(calculate_age)
```

Figure B.4: Deriving the Age,



4.1.4. Improvise new feature, team's Form Indicator

```
# Establish the Form Indicator of the teams based on Age derived
fb_df.sort_values('Age', inplace=True)

# Create a new column to store the form indicator for each team
fb_df['FormIndicator'] = 0

# Iterate over each raw in the data frame
for index, row in fb_df.iterrows():
    home_team = row['HomeTeam']
    away_team = row['AwayTeam']

    # Get the previous 5 matches  for the home and away teams
    previous_home_matches =fb_df[(fb_df['HomeTeam'] == home_team) & (fb_df['Date'] < row['Date'])]
    previous_away_matches =fb_df[(fb_df['AwayTeam'] == away_team) & (fb_df['Date'] < row['Date'])]

    # Calculate the avarage points obtained by the home and away teams in their previous matches.
    home_team_points = previous_home_matches['FTR'].map({'H': 3, 'A': 0, 'D': 1}).sum()
    away_team_points = previous_away_matches['FTR'].map({'H': 3, 'A': 0, 'D': 1}).sum()

    # Update the form indicator for home and away teams
    fb_df.at[index, 'FormIndicator'] = home_team_points - away_team_points

# Print the updatad data frame with the form indicator
print(fb_df[['Age', 'HomeTeam', 'AwayTeam', 'FormIndicator']])
```

```
   Age        HomeTeam      AwayTeam  FormIndicator
,
     0         Torquay       Wrexham              6
     0           Derby     Morecambe             -7
     0          Exeter      Charlton             -7
     0  Fleetwood Town        Burton             -7
     0    Forest Green     Peterboro             -7
   ...             ...           ...            ...
    10    Crawley Town     Port Vale             -7
    10           Crewe     Stevenage             -7
    10      Gillingham  Notts County             -7
    10         Oldham       Carlisle             -7
    10        Arsenal   Aston Villa              7

i398 rows x 4 columns]
```

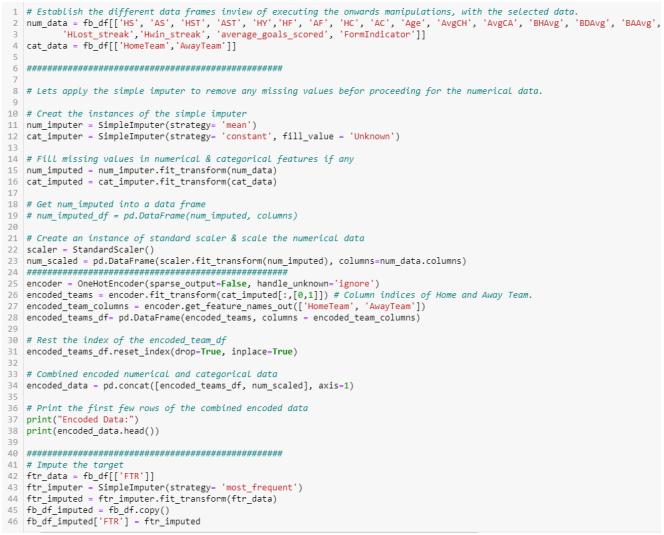
Figure B.5: Form Indicator,

```
             PC1        PC2        PC3        PC4       PC5       PC6  \
0      48.278472 -20.147435   5.014671  -3.752919 -1.385801  0.058082
1      68.037080  22.324624   8.500066  15.296915  2.632389  1.046336
2      60.226217   1.711378  -1.704181   7.887853  0.227384 -0.617860
3      51.903442  -9.345940  -3.049383   4.250364  7.016374 -3.938427
4      72.522588  -3.078842  -9.716996   2.241886 -3.578629  3.560607
...          ...        ...        ...        ...       ...       ...
25393 -50.284624   0.910898  -2.842631   6.425929  7.348318 -4.114665
25394 -52.526502  -0.138733 -10.781674   3.438919 -3.467752  1.189621
25395 -53.589146   1.848403  -6.672797   3.593832  7.693655 -1.447447
25396 -53.046699   1.638664  -3.029440   7.930891  1.704802 -1.119756
25397 -54.761174   0.685052   5.259994  -2.023364 -1.815390  9.223880

            PC7       PC8       PC9      PC10  A  D  H
0      0.309314  0.286294  0.213641 -2.434161  0  1  0
1      5.953116 -4.542022  1.486426  4.459570  0  0  1
2      4.371249  7.476981  2.295193 -0.443490  1  0  0
3     -0.325362  1.068011 -3.586778  0.296642  1  0  0
4     -4.100190  3.852706 -0.847972 -1.070413  1  0  0
...         ...       ...       ...       ... .. .. ..
25393  5.185915  3.216514 -1.514212 -0.745773  1  0  0
25394  5.391202  4.597340  1.098013  1.326950  1  0  0
25395 -1.706015 -4.285512  0.275082  0.840733  0  0  1
25396 -3.781856  0.580220  3.477878  0.044807  0  0  1
25397  0.577693 -0.178158 -2.013496  0.706244  1  0  0

[25398 rows x 13 columns]
```

Figure B.6: Principle Components,



```
# Establish the different data frames inview of executing the onwards manipulations, with the selected data.
num_data = fb_df[['HS', 'AS', 'HST', 'AST', 'HY','HF', 'AF', 'HC', 'AC', 'Age', 'AvgCH', 'AvgCA', 'BHAvg', 'BDAvg', 'BAAvg',
       'HLost_streak','Hwin_streak', 'average_goals_scored', 'FormIndicator']]
cat_data = fb_df[['HomeTeam','AwayTeam']]

##################################################

# Lets apply the simple imputer to remove any missing values befor proceeding for the numerical data.

# Creat the instances of the simple imputer
num_imputer = SimpleImputer(strategy= 'mean')
cat_imputer = SimpleImputer(strategy= 'constant', fill_value = 'Unknown')

# Fill missing values in numerical & categorical features if any
num_imputed = num_imputer.fit_transform(num_data)
cat_imputed = cat_imputer.fit_transform(cat_data)

# Get num_imputed into a data frame
# num_imputed_df = pd.DataFrame(num_imputed, columns)

# Create an instance of standard scaler & scale the numerical data
scaler = StandardScaler()
num_scaled = pd.DataFrame(scaler.fit_transform(num_imputed), columns=num_data.columns)
##########################################
encoder = OneHotEncoder(sparse_output=False, handle_unknown='ignore')
encoded_teams = encoder.fit_transform(cat_imputed[:,[0,1]]) # Column indices of Home and Away Team.
encoded_team_columns = encoder.get_feature_names_out(['HomeTeam', 'AwayTeam'])
encoded_teams_df= pd.DataFrame(encoded_teams, columns = encoded_team_columns)

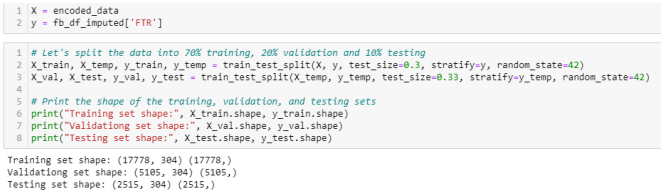
# Rest the index of the encoded_team_df
encoded_teams_df.reset_index(drop=True, inplace=True)

# Combined encoded numerical and categorical data
encoded_data = pd.concat([encoded_teams_df, num_scaled], axis=1)

# Print the first few rows of the combined encoded data
print("Encoded Data:")
print(encoded_data.head())

##################################################
# Impute the target
ftr_data = fb_df[['FTR']]
ftr_imputer = SimpleImputer(strategy= 'most_frequent')
ftr_imputed = ftr_imputer.fit_transform(ftr_data)
fb_df_imputed = fb_df.copy()
fb_df_imputed['FTR'] = ftr_imputed
```

Figure B.7: Establish Data Frames,



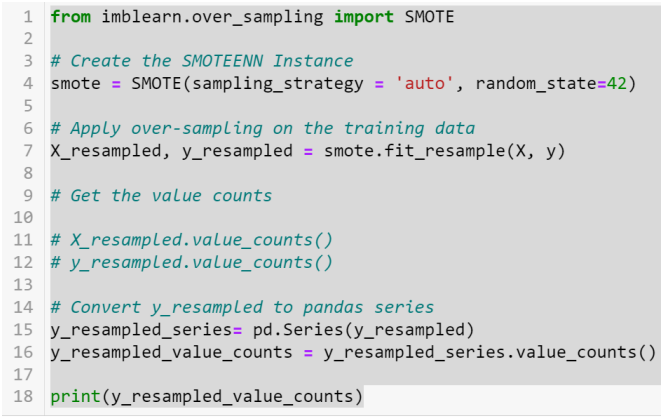
```
X = encoded_data
y = fb_df_imputed['FTR']
```

```
# Let's split the data into 70% training, 20% validation and 10% testing
X_train, X_temp, y_train, y_temp = train_test_split(X, y, test_size=0.3, stratify=y, random_state=42)
X_val, X_test, y_val, y_test = train_test_split(X_temp, y_temp, test_size=0.33, stratify=y_temp, random_state=42)

# Print the shape of the training, validation, and testing sets
print("Training set shape:", X_train.shape, y_train.shape)
print("Validationg set shape:", X_val.shape, y_val.shape)
print("Testing set shape:", X_test.shape, y_test.shape)
```

```
Training set shape: (17778, 304) (17778,)
Validationg set shape: (5105, 304) (5105,)
Testing set shape: (2515, 304) (2515,)
```

Figure B.8: Data splitting,



```
from imblearn.over_sampling import SMOTE

# Create the SMOTEENN Instance
smote = SMOTE(sampling_strategy = 'auto', random_state=42)

# Apply over-sampling on the training data
X_resampled, y_resampled = smote.fit_resample(X, y)

# Get the value counts

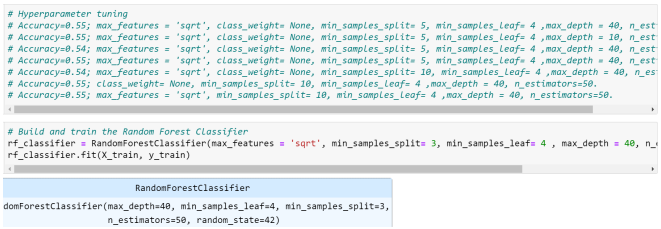
# X_resampled.value_counts()
# y_resampled.value_counts()

# Convert y_resampled to pandas series
y_resampled_series= pd.Series(y_resampled)
y_resampled_value_counts = y_resampled_series.value_counts()

print(y_resampled_value_counts)
```

```
D    10912
H    10912
A    10912
Name: FTR, dtype: int64
```

Figure B.9: Data resampled with SMOTE,



```
# Hyperparameter tuning
# Accuracy=0.55; max_features = 'sqrt', class_weight= None, min_samples_split= 5, min_samples_leaf= 4 ,max_depth = 40, n_est
# Accuracy=0.55; max_features = 'sqrt', class_weight= None, min_samples_split= 5, min_samples_leaf= 4 ,max_depth = 10, n_est
# Accuracy=0.54; max_features = 'sqrt', class_weight= None, min_samples_split= 5, min_samples_leaf= 4 ,max_depth = 40, n_est
# Accuracy=0.55; max_features = 'sqrt', class_weight= None, min_samples_split= 5, min_samples_leaf= 4 ,max_depth = 40, n_est
# Accuracy=0.54; max_features = 'sqrt', class_weight= None, min_samples_split= 10, min_samples_leaf= 4 ,max_depth = 40, n_es
# Accuracy=0.55; class_weight= None, min_samples_split= 10, min_samples_leaf= 4 ,max_depth = 40, n_estimators=50.
# Accuracy=0.55; max_features = 'sqrt', min_samples_split= 10, min_samples_leaf= 4 ,max_depth = 40, n_estimators=50.

# Build and train the Random Forest Classifier
rf_classifier = RandomForestClassifier(max_features = 'sqrt', min_samples_split= 3, min_samples_leaf= 4 , max_depth = 40, n_
rf_classifier.fit(X_train, y_train)

RandomForestClassifier
domForestClassifier(max_depth=40, min_samples_leaf=4, min_samples_split=3,
                    n_estimators=50, random_state=42)
```

Figure B.10: Hypeparameter Testing RFC,

```
# hyperparameter tuning
# Accuracy: 0.54: n_neighbors= 60, weights= 'distance', p=2
# Accuracy: 0.58: n_neighbors= 40, weights= 'distance', p=2
# Accuracy: 0.58: n_neighbors= 20, weights= 'distance', p=2
# Accuracy: 0.53: n_neighbors= 60, weights= 'distance', p=2
```

Figure B.11: knn Hyperparameter Tuning,

```
# For tuning hyper parameters, define the grid
gbc_param_grid = {
    'n_estimators': [80, 100,],   # Number of trees in the forest
    'learning_rate':[0.1],
    #'max_depth' : [3,8]    # The max depth of trees
    #'min_samples_split': [2,5,10],  # Max no of samples required to split internal node
    #'min_samples_leaf' : [1,2,4],   # Max no of samples required to be at a leaf node
    #'max_features': ['sqrt', 'log2', None], # No of features in looking fo the best split
    #'class_weight' : [None, 'balanced']       # weight to address class imbalance
}

# Let's creats the object of gridsearch  & Perform the Grid Search using training data
gbc_grid_search = GridSearchCV(gb_classifier, gbc_param_grid, cv=3, scoring= 'accuracy')
gbc_grid_search.fit(X_train, y_train)
```

Figure B.12: GBC Grid Search,

```
#Hyperparameter tuning
hidden_layer_sizes=(100,50), max_iter=300, Accuracy= 0.48
learning_rate= 'invscaling', alpha=0.0001, solver= 'adam', Accuracy= 0.48.4
activation ='tanh'. accuracy= 49.2
hidden_layer_sizes=(100,50), max_iter= 100, activation ='tanh'. Accuracy= .47
hidden_layer_sizes=(200,300), max_iter= 700, activation ='tanh'. Accuracy= .4884
hidden_layer_sizes=(100,50), max_iter= 500, activation ='tanh'. Accuracy= .4723
hidden_layer_sizes=(200,100), max_iter= 500, activation ='tanh'. Accuracy= .4846
hidden_layer_sizes=(200,300), max_iter= 500, activation ='tanh'. Accuracy= .4884
hidden_layer_sizes=(400,300), max_iter= 500, activation ='tanh'. Accuracy= .4688
hidden_layer_sizes=(300,300), max_iter= 500, activation ='tanh'. Accuracy= .4879
hidden_layer_sizes=(300,200), max_iter= 500, activation ='tanh'. Accuracy= .4738
hidden_layer_sizes=(activation ='tanh'). Accuracy= .4804
batch_size = 64, hidden_layer_sizes=(200,300), max_iter= 500, activation ='tanh'.Accuracy = .4809
batch_size = 32, hidden_layer_sizes=(200,300), max_iter= 500, activation ='tanh'.Accuracy = .4854
learning_rate= 'adaptive', hidden_layer_sizes=(200,300), max_iter= 500, activation ='tanh'.Accuracy = .4884
```

Figure B.13: Hyperparameter Tuning of MLP,

Appendix C

Link to Jupiter Notebook

hyperref

Link to Jupiter Notebook

Chathura Gamage

September 29, 2023

C.1 https://drive.google.com/file/d/1wYkU2lJffc view?usp=sharing